

Analog IC Design Implementation and Verification

Week 11 – Floorplanning, Placement and Routing

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Floorplanning

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Module Objectives

In this module, you learn how to

- Estimate the size of an analog-on-top chip given basic parameters.
- Identify the problems (if any) in a proposed floorplan.



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Floorplan: Introduction

Floorplan is a representation of the overall layout of an integrated circuit at a higher level of abstraction.

- Process of floorplanning consists of creating a sketch of the layout. The sketch shows the potential placement of bond pads and the location and shapes of the cells.
- When the layout of most or all of the cells is complete, the floorplan can be used as a template to assemble the top-level layout.
- A good floorplan leads to a good layout with few iterations.
- The information required to construct a floorplan includes
 - List of all the pads
 - Order and placement of all the pads
 - Estimated area of individual cells
 - Estimated area of the die



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Floor Plan Factors

A high quality floorplan with a complex hierarchical design ensures high speed and performance.

- The quality of the floorplan determines the quality of final design.
- Poor floorplanning can cause the failure of the fabricated integrated circuit (IC) to meet issues such as signal integrity, timing closure, routing congestion, high power dissipation, higher area, and huge IR drop.



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Floor Plan Factors (continued)

Floorplanning involves the following activities.

- Decision on pin and pad location
- Considering package constraints and power, ground pad locations
- Area estimation of blocks
- Decision on block placement
 - Considering pad locations
 - Considering timing/IR drop and signal integrity (SI) effects
- Decisions on routing/placement blockage
- Location and area of the soft blocks and their pin locations
- Number of power pads and their locations



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Block Size Estimation

It is very important for a layout designer to estimate the size and structure of a block, in addition to designing the individual components and how to interconnect them.

- Common practice among floorplanners is to use spread sheet calculations or paper-and-pen exercises.
- New electronic design automation (EDA) tools are becoming available that predict the size of the chip during the initial phases of the design cycle.

For example, one tool is ChipEstimate automatic layout size estimator.

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1. Talk about ChipEstimate, automatic layout size estimator.

2. Utilization factor:

Ex: If Utilization factor = 0.65

This mean that 65% of the total area of the block is used for device placement and the remaining 35% can be used for routing and other purposes.

Block Size Estimation (continued)

- Total block size is the sum of the cell size plus area required for routing.
- The ratio between the total area of the block and the total area consumed by the devices, is called *utilization factor* of a block.

Utilization factor = area (placement) consumed by the devices / total area of the block

- Details of block-level size and area estimation are covered in the subsections that follow.



1. Talk about ChipEstimate, automatic layout size estimator.

2. Utilization factor:

Ex: If Utilization factor = 0.65

This mean that 65% of the total area of the block is used for device placement and the remaining 35% can be used for routing and other purposes.

Resistance Area Estimation

The area A required to construct one or more resistors can be estimated with this formula:

$$A = 1.2 \cdot R \cdot W_r \cdot (W_r + S_r) / R_s$$

where:

- R = desired resistance
- R_s = applicable sheet resistance
- W_r = width of the resistor
- S_r = spacing between adjacent resistor stripes

The factor 1.2 accounts for the space consumed by dummy resistors, contact heads, and unused space in the layouts.

Note: Resistors of different widths or different materials must be computed separately.



For example, 122 kohm HSR with a width of 6 μm and a spacing of 12 μm consumes an estimated 7900 μm² of die area.

Resistor Area Calculation: Poly Resistor

Resistor area can be estimated by using the equation:

$$A = 1.2 \cdot R \cdot W_r \cdot (W_r + S_r) / R_s$$

where:

$$R = 537.33 \text{ ohms}$$

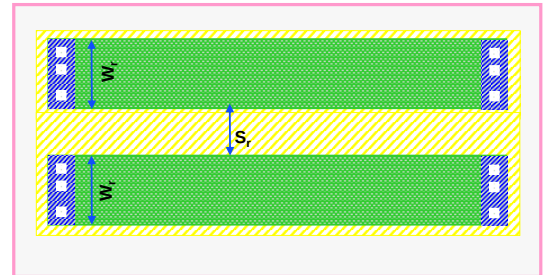
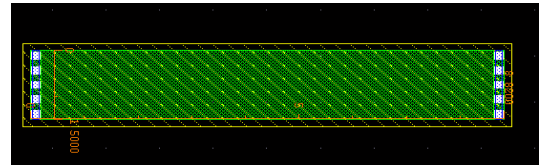
$$R_s = 100 \text{ ohms}$$

$$W_r = 1.5 \text{ } \mu\text{m}$$

$$S_r = 0.3 \text{ } \mu\text{m}$$

Area calculated from the layout = $13.8 \text{ } \mu\text{m}^2$

Area estimated from the equation = $17.3 \text{ } \mu\text{m}^2$



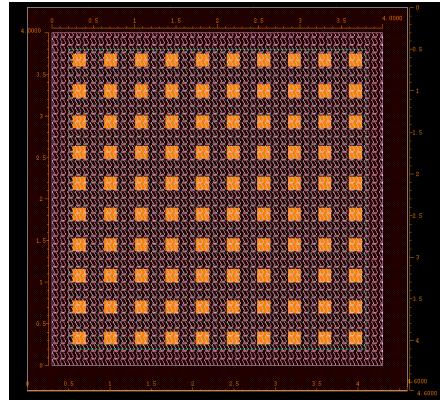
Note: The actual area of the poly resistor is calculated from the GPDK reference.

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Capacitor Area Estimation

The area required by capacitors depends on the capacitance per unit area of the dielectric.

- For example, capacitance per unit area of a metal-insulator-metal capacitor (MIMCAP) is $1.1 \text{ fF}/\mu\text{m}^2$.
- Estimated area of a 17.6 fF MIMCAP is $16 \mu\text{m}^2$.
- Area estimated from theory is $16 \mu\text{m}^2$.
- Area calculated from layout is $21.6 \mu\text{m}^2$.



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MOS Transistors Area Estimation

The area A required for a finger-style MOS transistor can be estimated by:

$$A = 1.3 \cdot W_g \cdot (L_g + S_{gg})$$

where:

W_g	=	width of the gate
L_g	=	length of the gate
S_{gg}	=	spacing between adjacent gate stripes of a multiple-finger transistor

The factor 1.3 accounts for the space consumed by the terminations on either end of the transistor array, well spacing, and less-than-ideal packing.



This formula usually underestimates the area required by small transistors, particularly if they require guard rings or separate wells.

To get the correlation, fingers greater than 10 need to be used.

MOS Transistor Area Calculation

Area estimated by using the equation:

$$A = 1.3 W_g (L_g + S_{gg})$$

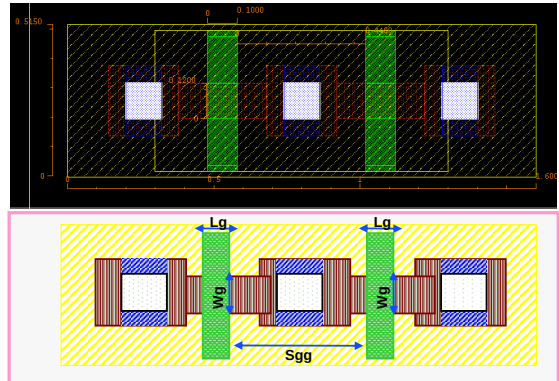
where:

$$L_g = 0.1 \mu\text{m}$$

$$W_g = 0.12 \mu\text{m}$$

$$S_{gg} = 0.44 \mu\text{m}$$

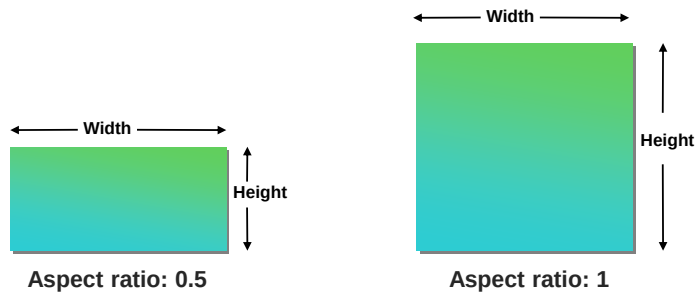
- Area estimated by the equation is $0.084 \mu\text{m}^2$.
- Area calculated from layout is $0.859 \mu\text{m}^2$.
- In general, correlation happens when transistors with a larger area are taken into account (fingers > 10).



For small transistors, the correlation doesn't happen.

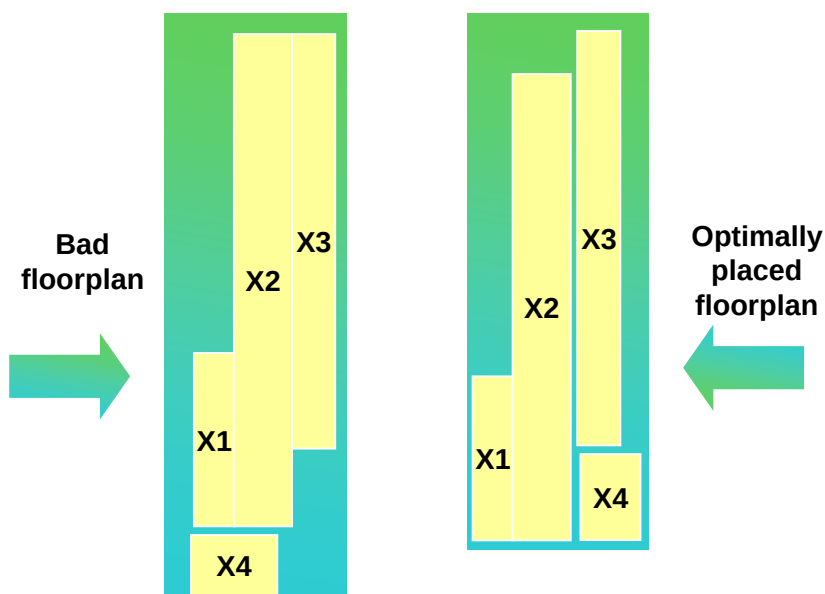
Aspect Ratio for a Block

Aspect ratio in floorplanning is the physical shape of a block, which is often represented as height/width (H/W).



Aspect ratio of a block

Floor Planning Techniques: Aspect Ratio



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By changing the aspect ratio of instance X4 from 0.5 to 2 for the above said example, we were able to reduce the overall height of the block significantly.

Chip Area Estimation

The first phase of chip area estimation consists of compiling a list of all the cells including full custom blocks and routed areas used in the design. This list includes power devices that require any specific locations due to matching or routing considerations.

- Some cells already might have been laid out for previous designs, otherwise known as *hard blocks*. In that case, area estimates for these cells can be easily obtained.
- If no previous layout exists, then the area of the cell must be computed from the areas of the individual components, otherwise known as *soft blocks*.
- Area estimates are usually given in either square millimeters (mm^2) or in thousands of square mils (kmi^2), where $1 \text{ kmi}^2 = 0.645 \text{ mm}^2$ and $1 \text{ mm}^2 = 1.55 \text{ kmi}^2$.
- Chip area estimation is the sum of the estimated areas of the individual cells present in the block.



In custom analog circuits, the cell area estimation can be attained only through a thorough understanding of all schematic blocks/components.

If detailed schematics are available, then this task amounts a little more than listing the cells found in the top-level schematic. If no schematic exist, then the circuit designer must prepare a list based on a careful examination of the specifications. The list should only contain cells appearing in the top level schematic, and should exclude cells occupying the lower-level cells of schematic hierarchy.

Chip Size

It is very important for a floorplanner to predict the size and structure of a die accurately, because the size of the die directly impacts the cost of the product.

- ASIC vendors and designers currently estimate chip size from their past experiences using spread sheet calculations.
- EDA tools are becoming available that predict the size of the chip during the first phases of the design cycle.

For example, ChipEstimate makes tools that automatically estimate chip size.



Also Form Factor.

Talk about ChipEstimate, automatic layout size estimator.

Computing Chip Area

The area of a chip A_{chip} can be estimated with the following formula:

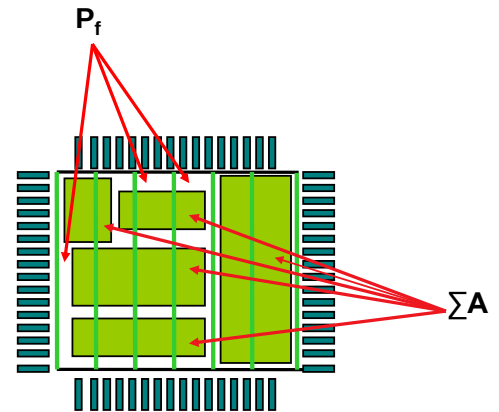
$$A_{\text{chip}} = P_f \sum A$$

where:

$\sum A$ = Sum of the areas of the individual components

P_f = Packing factor

Note: The packing factor accounts for the area consumed by isolation and interconnection and the area used by imperfect packing.



Standard bipolar designs employing single-level metal typically have packing factors of 1.5 to 3.0.

Values toward the lower end of this range represent well-packed designs using custom-crafted devices and extensive device managers.

Values toward the upper end of this range represent designs using standardized components and few or no device mergers.

Die Size Estimation

Some factors considered for die-size estimation include the following:

- List of IPs and their area
- Target manufacturing process
- If the chip is pad limited or core limited
- Total area of the memories
- Package requirement (flip chip or wire-bond chip)
- Utilization factor
- Pad ring preparation details
- Approximate gate count/mm²
- Total chip area
- Area for scribe line



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Area Estimation

Core area can be estimated by using the equation:

$$A_c = R_f \cdot P_f \cdot \sum A_{\text{cell}} + P_f \cdot \sum A_{\text{pwr}}$$

where:

$\sum A_{\text{cell}}$ = sum of all the cell areas

$\sum A_{\text{pwr}}$ = sum of areas of all power devices not contained in any cell

R_f = routing factor accounts for the area consumed by top level wiring

P_f = packing factor represents area wasted between cells. (Hand-optimized designs have $P_f = 1$.)



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Area Estimation (continued)

Die area can be estimated with the equation:

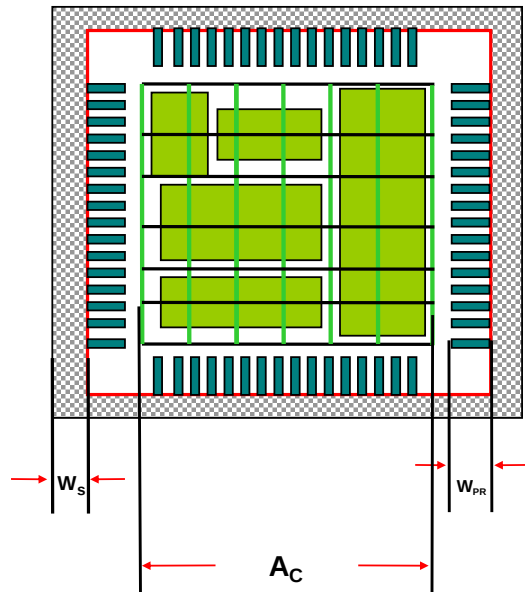
$$A_{\text{die}} = (\sqrt{A_c} + (2 W_{\text{pr}} + W_s))^2$$

where:

W_{pr} = width of the pad ring

W_s = scribe width

A_c = area of the core



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Area Estimation (continued)

Die perimeter can be estimated using:

$$P_{\min(\text{approx})} = (N_p + 4)(W_p + S_p) + 4W_s$$

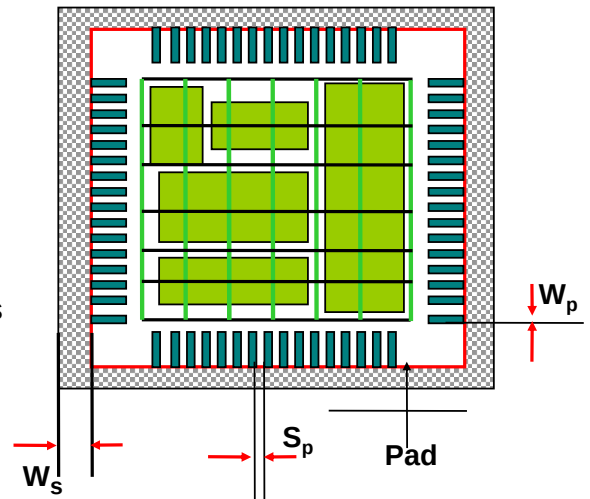
where:

N_p = total number of pads

W_p = width of the pad

W_s = width of the scribe

S_p = minimum space between adjacent pads



W_{pr} = Width of the padding is usually 130% width of bond pads.

W_s = scribe width usually (75 to 125 microns).

Core Limited or Pad Limited?

Perimeter utilization factor (P) is the fraction of the available pad perimeter used by the pads.

Assuming the aspect ratio is square:

$$P = \frac{P_{\min}}{4\sqrt{A_{die}}}$$

- If $P < 1$, design is core limited.
- If $P > 1$, design is pad limited.



Draw a square with $p/4$ as side

$A_{die} = p^2/16$

$P = 4\sqrt{A_{die}}$

Core Limited or Pad Limited? (continued)

What if the chip is core limited?

- Migrate to a smaller process technology.

What if the chip is pad limited?

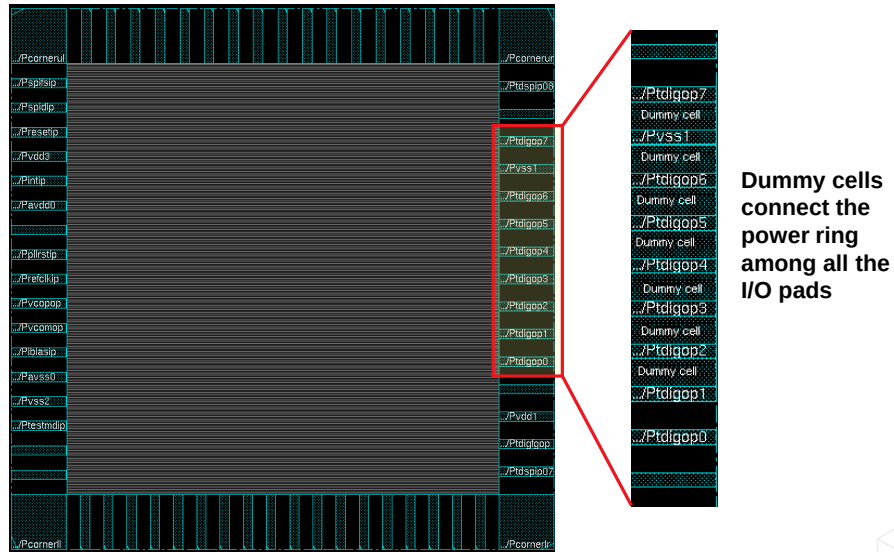
- Try to share the pins. Each pin then has dual function, configured by chip.
- Look at flip chip possibility. You can put multiple rows of pads.
- Minimize the logic and area.
- Remove functions that are not necessary.



All these have to be considered during the planning stage,
Using the eqns.. Given for area calc.. In the previous slides.

Core-Limited Design

In *core-limited design*, the core packs tightly into the space inside pad ring without enough pads to free the ring.



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Talk about typical high gate count designs where the area of the core dominates and the I/O pads are sparsely spaced,

To have the PAD ring continue these sparsely spaced I/O pads are inserted with dummy pads which enable the power rings to continue between these separated I/O pads.

Chip Aspect Ratio

The aspect ratio of a chip is decided after considering several design parameters:

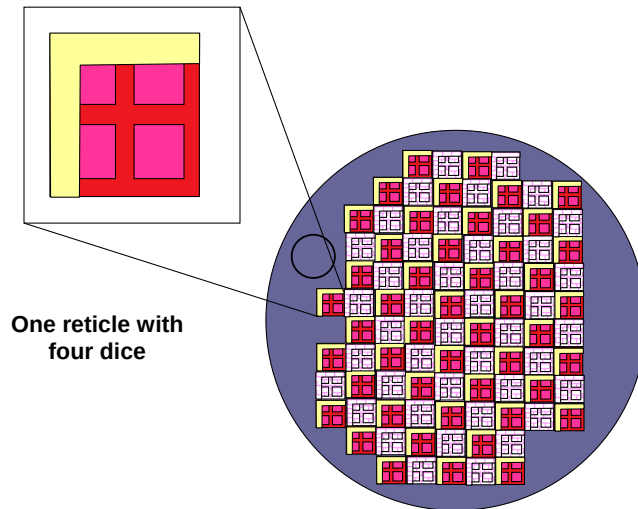
- Process node (for example, 65 nm or 90 nm)
- Wafer size (200 mm or 300 mm)
- Shape or size of major IPs
- Reticle limit (number of dice exposed per exposure)



Wafer size and process nodes are an important factor for any design/project manager to decide upon the shape and size of a chip during the project planning stage.

Reticle Limit

One reticle in a mask plate is stepped and repeated across a wafer.



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A **photomask** is an opaque plate with holes or transparencies that allow light to shine through in a defined pattern. They are commonly used in photolithography. Lithographic photomasks are typically transparent fused silica blanks covered with a pattern defined with a chrome metal absorbing film. Photomasks are used at wavelengths of 365 nm, 248 nm, and 193 nm. Photomasks have also been developed for other forms of radiation such as 157 nm, 13.5 nm (EUV), X-ray and electrons and ions, but these require entirely new materials for the substrate and the pattern film.

A set of photomasks, each defining a pattern layer in integrated circuit fabrication, is fed into a photolithography stepper or scanner and individually selected for exposure. In double patterning techniques, a photomask would correspond to a subset of the layer pattern.

In photolithography for the mass production of integrated circuit devices, the more correct term is usually **photoreticle** or simply **reticle**. In the case of a photomask, there is a one-to-one correspondence between the mask pattern and the wafer pattern. This was the standard for the 1:1 mask aligners that were succeeded by steppers and scanners with reduction optics. As used in steppers and scanners, the reticle commonly contains only one layer of the chip. (However, some photolithography fabs utilize reticles with more than one layer patterned onto the same mask). The pattern is projected and shrunk by four or five times onto the wafer surface. To achieve complete wafer coverage, the wafer is repeatedly 'stepped' from position to position under the optical column until full exposure is achieved.



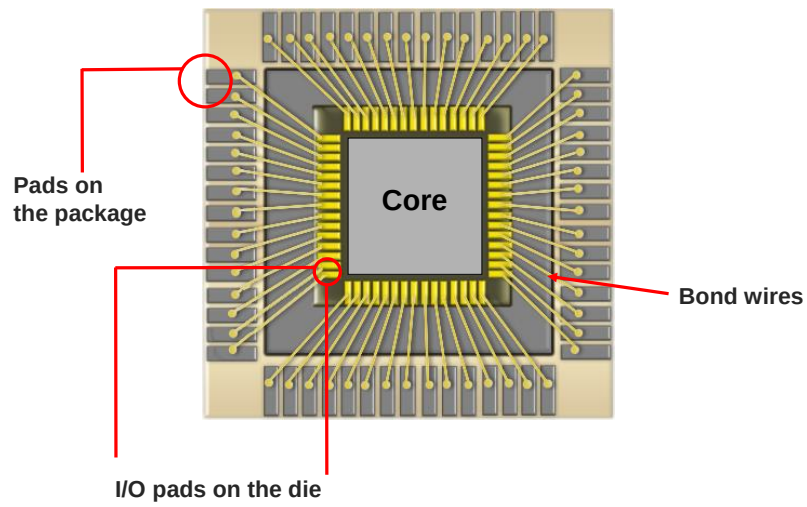
Subsection

Location of Bond Pads and Influence of Bond Wires

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Bond I/O Pad

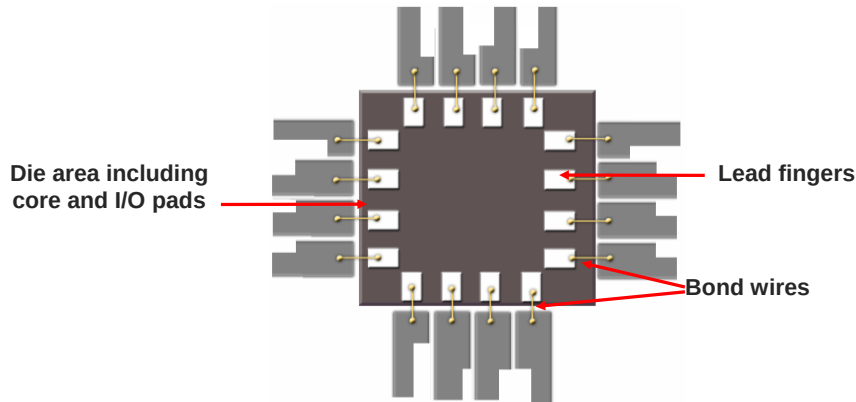


Wire bond chip.

Location of Bond Pads

The primary or fundamental requirement for the placement of bond pads is that the floorplan has to be superimposed on a copy of the lead frame drawing.

Initially the bond pads are made to sit just next to their lead frame fingers. This arrangement results in the shortest bond wires and the largest separation between the bond wires.



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Lead frame is a common arrangement in which the lead fingers actually encircle the mount pad. Once the dimensions of the die have been determined, a lead frame must be chosen.

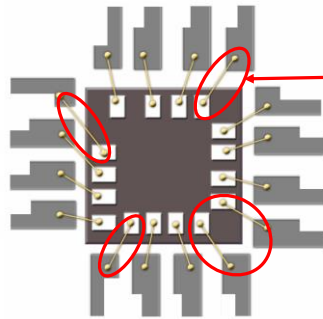
A chip is attached in the center of the substrate, and physical wiring connections made between the chip and the substrate via discrete wires. Chips have an array of bond pads around the perimeter to which these discrete wires are attached. An analogous bond pad on the substrate is the wire bond finger, referred to as fingers because they are usually long and narrow and fan out like the fingers of an outstretched hand. Thus, wire bond fingers are a series of I/O pads physically resident on a chip carrier (lead frame) and which connect to I/O pads on a chip. When the chip is placed within the cavity of a chip carrier, wires run out radially from the chip and are attached to the wire bond pads.

A unique set of wire bond fingers is generated for each substrate/chip package. The wire bond fingers must be small enough to squeeze into the space available but large enough to meet packaging ground rules. The fingers must be placed at the appropriate angle (i.e., fan out or fan in) in order to support the printed wiring that is attached on the substrate surface. The fingers must be spaced far enough apart to allow the wire bonder head to move past a finger without dislodging a discrete wire from on an adjacent finger. And each finger must meet minimum and maximum spacing guidelines from the chip as well as the surrounding substrate wiring.

Bad Location of Bond Pads

However in the previous arrangement the interconnect length between the pads and circuit blocks (internal to the core) can be affected.

- As a remedy, the pads can be moved slightly to accommodate the layout. If the pads are moved too far, the design cannot be manufactured, as shown in the figure.
- The pads must have a minimum size and spacing in order to allow automated wire bonding machines physically attach wires during the manufacturing process.



Non-uniform placement of bond wires complicates the automated bonding activity.

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Controlling the Wirebonding Process

Once a smart package and a smart die have been obtained, an engineer needs to place the die into the package and bond it up. Some directions must be given to the bonding software. Depending on the designer's requirements, there are different modes available:

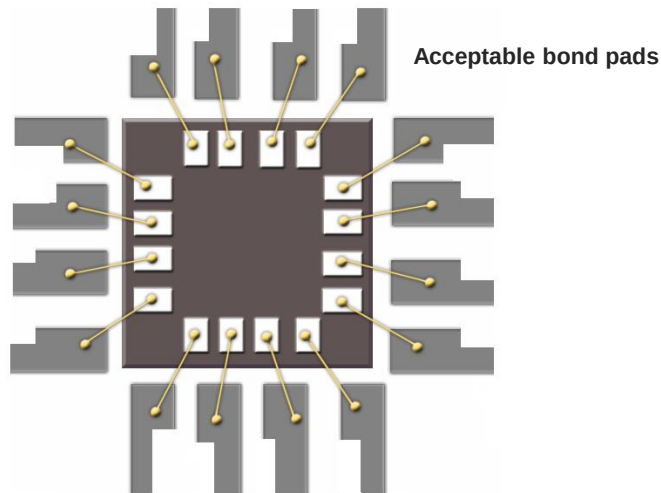
AutoBond: Allows the wirebond program to decide which die pads go to which fingers based on assembly rules (i.e., the wires out should go out as directly as possible to the finger, no wires should cross and no wires should be too close together).

Connection List: Defines exactly which die pads go to which package fingers. In this case, the assembly engineer provides a list to the program - mapping each die pad to each finger.

Net Rules: Some die pad destinations are fixed; others are free to bond where possible. This is often the case when die input/output nets go to fingers, and power and ground nets go to rings on the package.

Good Location of Bond Pads

The figure shows an acceptable bonding arrangement. This arrangement overcomes all the previous mentioned violations and also reduces the interconnect length between the bond pads and circuit blocks.



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Compared to BAD, the pads are placed evenly and the routing is feasible.

Rules for Bond Wires

There are some design rules for bond wires.

- There is a minimum and maximum wire length for the wire that attaches the chip I/O to the wire bond finger.
- The fingers also must angle properly so that they run along the axis of a straight line connecting the finger and the chip I/O pad.

For example, if the chip I/O is a voltage net (instead of a signal net), then the wire connection ends at a voltage ring.

- In order to allow the automated bonding, the bond wires must not intersect.
- The bond wires must not be close together.



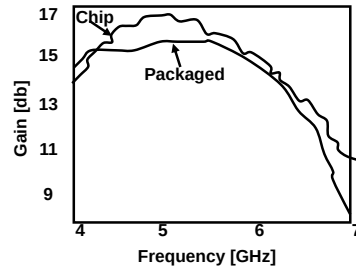
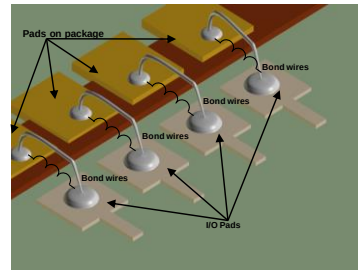
These are some of the rules followed for bond-wires.

Influence of Bond Wires

Bond wires connect a die to the pins of the package that holds it. These wires introduce relatively large inductance that results in crosstalk and signal distortion.

Let's consider an example of a chip that contains a critical low noise amplifier (LNA).

- The bond wire inductance can effect the return loss, gain, and noise figure of the amplifier.
- Overall this flattens the gain of the circuit due to the additional ground inductance from the ground bond wires.



Gain vs, frequency response of LNA

LNA -> Low noise amplifier.

What Is Flip-Chip?

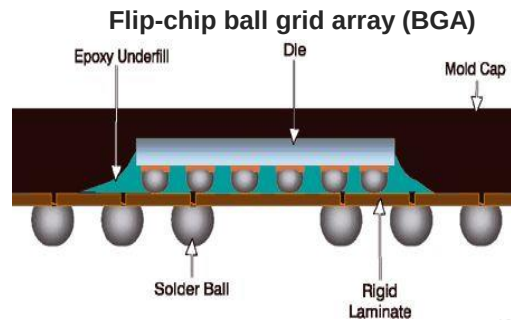
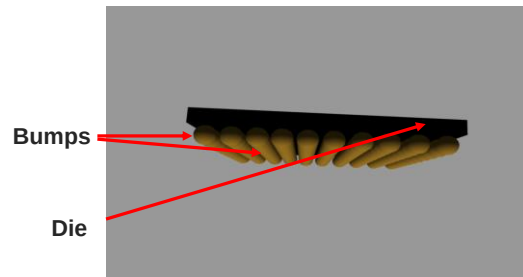
- Flip-chip is a method to electrically connect the die to the package carrier.
 - Lead frame or substrate.
- The bond wire is replaced with a conductive bump placed directly on the die surface.
 - Under-fill epoxy is used to secure the attachment and absorb stress.
- The chip is then flipped face down onto the package carrier using a re-flow process.
- Bump sizes range from 90 to 125 microns in diameter.
 - Height ranges from 60 to 100 microns.
- Also known as C4 (controlled collapsible compression connection).
 - Invented by IBM®.



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Flip-Chip Advantages

- Reduced signal inductance
 - Shorter interconnect lengths
- More efficient power use
 - Power directly at the core
- Higher interconnect density
 - More routable area
- Smaller package size
 - Chip scale packaging (CSP)
- I/O not controlling core size
 - (Core) area I/O placement
 - Possible die shrink



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Source of the picture:

Cadence Internal: The Advantages of Flip-Chip Design: John Park, February 2003

C4 controlled collapse chip connection

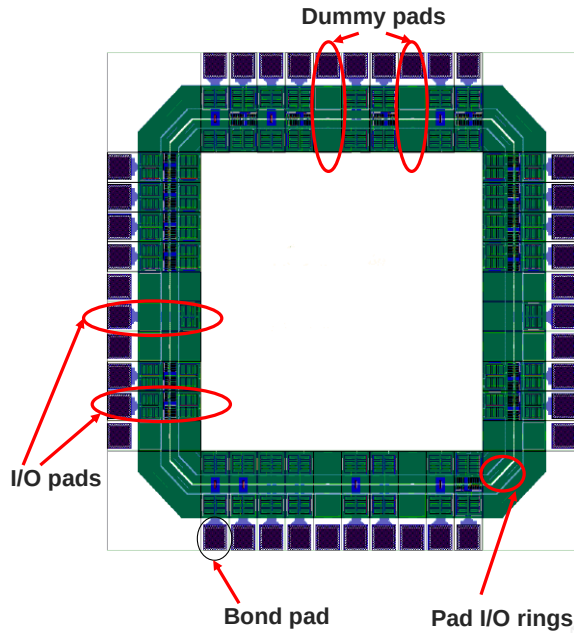
Notes on CSP

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Pad Ring

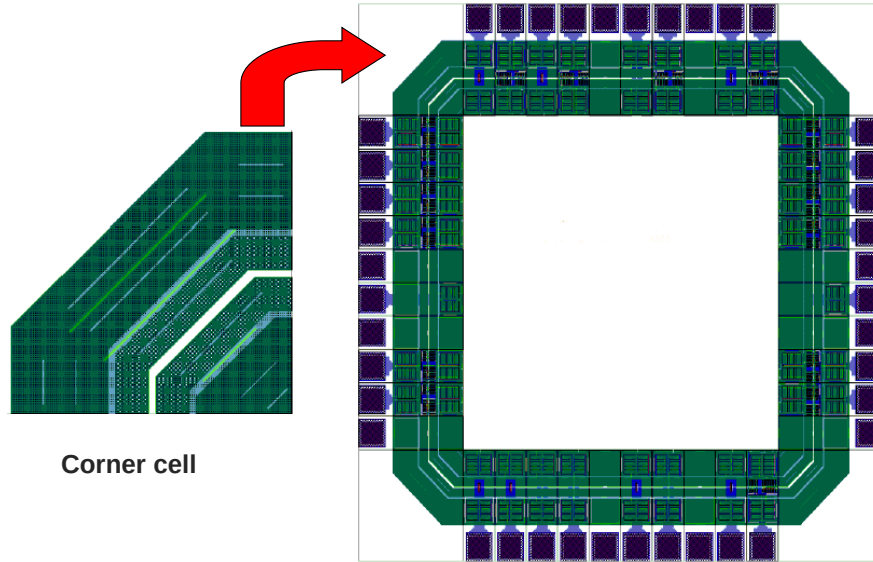
Pad ring is an integral part of an I/O pad.

- Pad rings cause a uniform power distribution throughout the die that connects several power pads.
- During a core-limited chip design, dummy pads are inserted between I/O pads to make the pad ring run throughout the periphery.
- Pad rings are drawn with highest possible widths to reduce IR drop on power lines.



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I/O Placement: Traditional Pad Frame



As we don't have I/O pads on the corners, a separate corner cell is inserted which connected all the pad rings.



Subsection

Separating Digital from Analog Areas

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Separation of Noisy Digital and Sensitive Analog Areas

Noise injection from digital to sensitive analog areas can happen through different mechanisms, such as these:

- Signal-to-signal coupling
 - Capacitive coupling
 - Inductive coupling
- Noise coupling through substrate

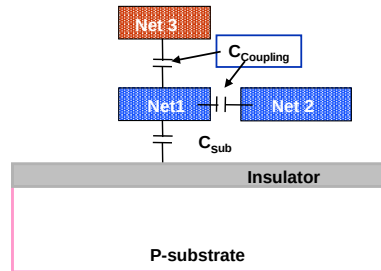


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Noise Injection Mechanisms: Capacitive

Signal-to-signal noise coupling: capacitive coupling

- When high-frequency digital signals (clocks and so forth) are placed next to sensitive analog signals (voltage reference and so forth), coupling happens through line-to-line parasitic capacitance from aggressor lines to victim lines.
- Clock rising or falling edges contain high frequency components. Parasitic capacitors are usually very low (usually in the order of fF) and respond to only high frequencies. Usually capacitive coupling happens during rising or falling edge.



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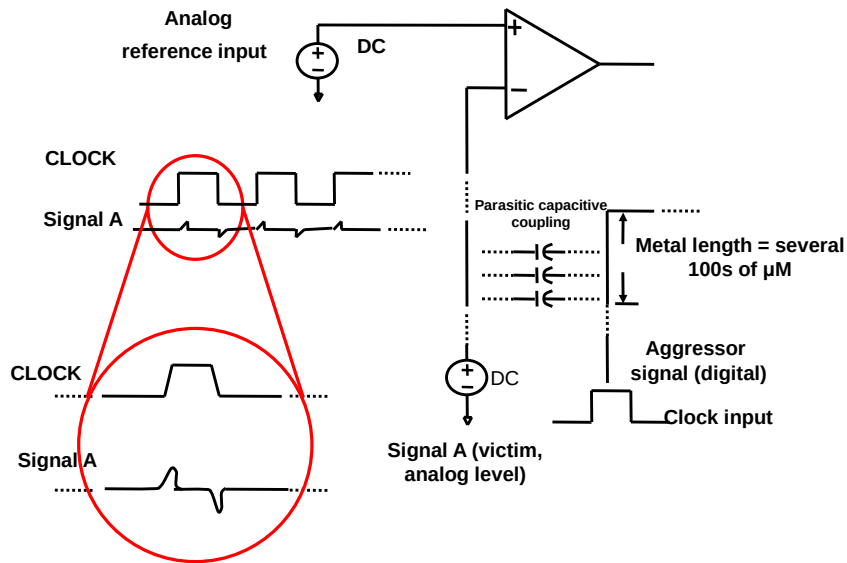


Coupling cap.. between aggressor and victim.

Next slide explains the effect of the coupling

Between a clock net and signal net.

Noise Injection Mechanisms: Capacitive (continued)



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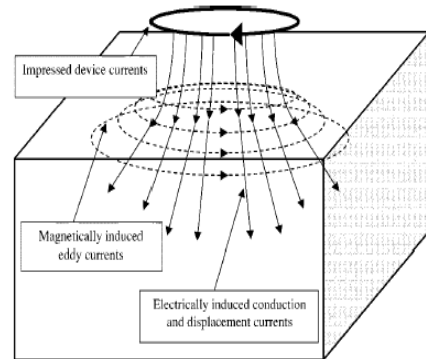


When a clock net and an analog DC ‘signal A’ net run for 100’s of meters together, we have the capacitive coupling between these nets and have the DC signal being influenced with small ripples due to high frequency raises and falls of clock net.

Noise Injection Mechanisms: Inductive

Signal-to-signal noise coupling: inductive coupling

- On-chip inductors have their own magnetic field which penetrates the substrate.
- Induced currents in the substrate run opposite of the current in the inductor loop.
- Substrate current opposes the original magnetic field, reducing stored magnetic energy in the inductor.
- Same process applies to solid metal plane located beneath the inductor.



* From A. M. Niknejad and R. G. Meyer, "Analysis of Eddy-Current Losses Over Conductive Substrates with Applications to Monolithic Inductors and Transformers," IEEE Trans. Microwave Theory Tech, vol. 49, pp. 166-176, Jan. 2001.

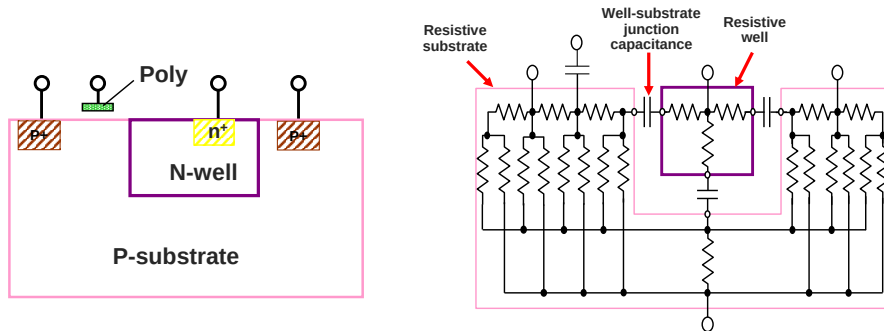
Source of the picture: Taken it from cadence Assura RF whitepaper.

* From A. M. Niknejad and R. G. Meyer, "Analysis of Eddy-Current Losses Over Conductive Substrates with Applications to Monolithic Inductors and Transformers," IEEE Trans. Microwave Theory Tech, vol. 49, pp. 166-176, Jan. 2001.

Noise Injection Mechanisms: Substrate

Substrate noise coupling

- Noise can couple from one end of a chip to the other, even if there are no wires. It can go right through substrate. A network of RC elements connect transistors to one another in substrate level.
- If analog circuit blocks are placed next to digital circuit blocks (logic or heavy-duty buffers switching at high frequency), noise can be injected into analog-circuit blocks through substrate.



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Millions of gates integrated on a chip can generate substrate potential fluctuation large enough to impair sensitive analog circuits on the same chip.

As a result, the substrate noise coupling can severely degrade the performance of sensitive circuitry.

A digital switching device induces currents to the substrate through the depletion capacitances of the device p-n junctions. The output buffers can be particularly noisy, as well as common-emitter bipolar transistors.

Due to the inductances associated to package pins:

noise at the power-supply lines (ΔI noise) is usually large in CMOS circuits.

These supply lines are typically used to bias the wells and substrate in the digital section. Thus the amount of noise introduced to the silicon bulk can be very large, all depending on the number and location of contacts and the ΔI noise level.

Also, this noise can forward-bias protection diodes, and introduce substrate currents.

As device feature sizes are being reduced:

impact-ionization currents have gained significance.

although traditionally the noise capacitively coupled through the junction is larger than impact-ionization, in deep submicron devices hot-carrier effects become a major source of noise injection

Injection of substrate noise:

Typical active devices include bipolar junction transistors, MOS transistors and diodes. Typical passive device include resistor, capacitors, inductors, and interconnects. Local wells and diffusions are belloyed to passive structures.

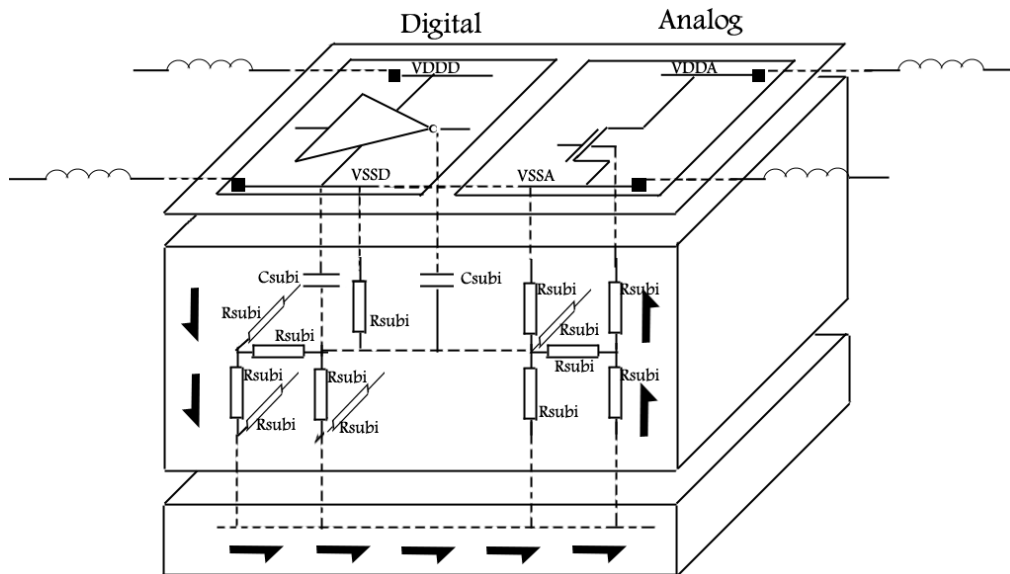
Reception of substrate noise

The reception of noise by most devices on the surface takes place through capacitive sensing. This is true of bipolar transistors, capacitors, resistors, and interconnect lines.

Transmission of substrate noise

Substrates act as the media for coupling of noise from one device to another

Noise Injection Mechanisms: Substrate (continued)



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This picture is a 3D representation of a chip which has both analog and digital blocks planned together,

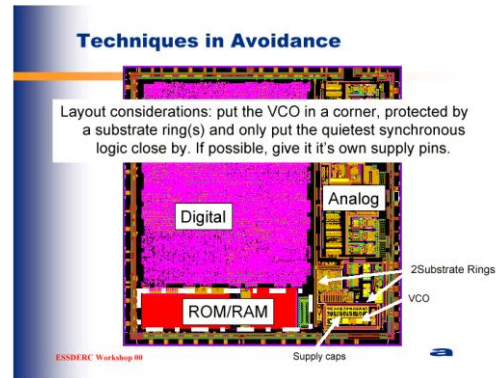
Though both analog and digital blocks are isolated on the layout view, they get electrically coupled through substrate parasitic due to which we can have the sensitive analog blocks get influenced with high frequency digital clocks. Overall this substrate coupling can affect the functionality of the chip.

There are tools which can extract the substrate parasitic and help in deduct digital block interference early in the design cycle. E.g. Assura RF

Substrate Noise-Related Failures: VCO

Situation: Video converter with 27 MHz PLL (phase locked loop) failed.

- PLL locked to the digital clock due to the substrate noise injection.
- The voltage-controlled oscillator (VCO) that was part of the PLL started responding to the spurious clock from the substrate and not its original clock.
- Later they found this noise injection was due to substrate noise coupling.
- Finding this flaw early in the design cycle could have saved the time spent on respins.



www.imse.cnm.es/esd-msd/WORKSHOPS/ESSDERC2000

Reference: www.imse.cnm.es/esd-msd/WORKSHOPS/ESSDERC2000

VCO : Voltage controlled oscillator.

Substrate disasters happen. This is an example that was presented at a technical conference a couple of years back. It is a video converter chip with an on-board 27Mhz PLL for the color burst. The problem was that the chip had striping at regular interval on the monitor. After long investigations the designers understood it was a substrate problem, with the VCO responding to the spurious “clock” that was from the substrate and not it’s original clock.

The designers did not use a substrate noise analysis tool, and therefore took a lot of different corrective actions: The VCO was placed in a corner away from all the noisy logic, a lot of guard banding was done, supplies were separated.

The result was that the chip worked, but nobody is able to tell which of the corrective actions was the good one. When this chip gets ported over to a newer technology, these same things will be done again, but there is no guarantee that it is going to work.

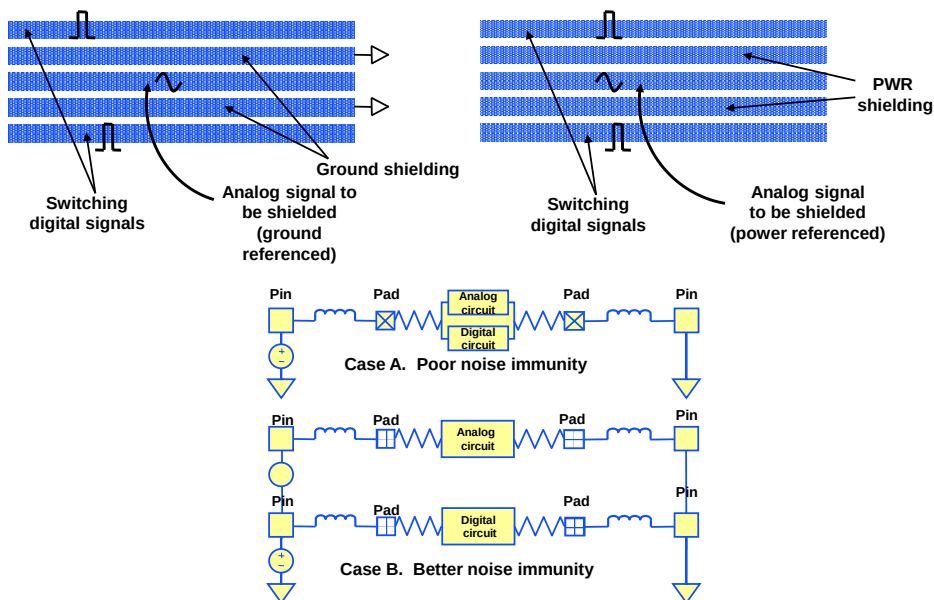
Layout Tips to Reduce Noise Coupling

- Use guard rings around all critical analog blocks and devices.
- Have separate analog and digital ground and power lines.
- Use inductors with higher metal layers to avoid magnetic interferences on the analog devices.
- Do not route over the high-power devices (restrict from M2 to top metal).
- Use minimal driver sizes (especially for digital circuits and buffers).
- Use distributed-decoupling capacitors to reduce power supply noise (high-speed).
- Increase power-supply-bus-routing width wherever possible to reduce inductance.



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Layout Tips to Reduce Noise Coupling



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Planning Hot Circuits

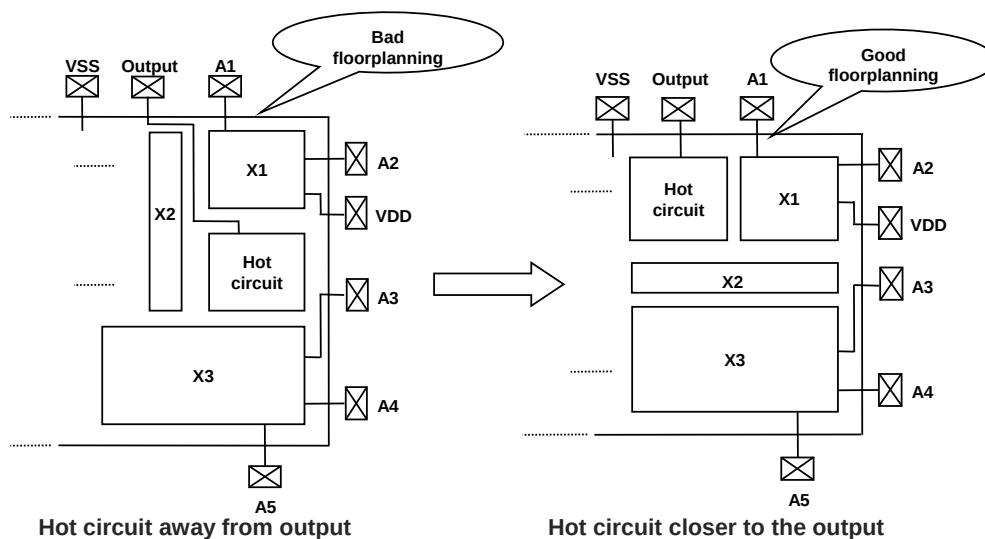
Hot circuits, such as LCD (liquid crystal diode) drivers or high power audio power amplifiers, can source or sink high output current, sometimes on the order of a couple of amps.

- This current can source from supply to output or sink from output to ground.
- The location of these hot circuits should be close enough to the output, supply, and ground pads.
- The width of metal connection from pad (output, supply and ground) to the device and the finger size of the hot device are some of the critical considerations in analog-on-top floorplanning.
- Good hot-circuit planning helps to have easier power planning.



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Example: Planning Hot Circuits



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Subsection

Planning Hot (and High-Power) Circuits

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Power and Ground Planning

Power and ground planning types:

- I/O cell power planning
- Core cell power planning
- I/O cell power planning addresses minimal resistance between VDD and VSS pad to each I/O pad.
- Core cell power planning is based on maximum power consumption of each cell.
 - For example, the output of a regulator is current limited.
 - Based on this limit, the power connection width and location of power and ground pins (with fixed floorplan) are selected.



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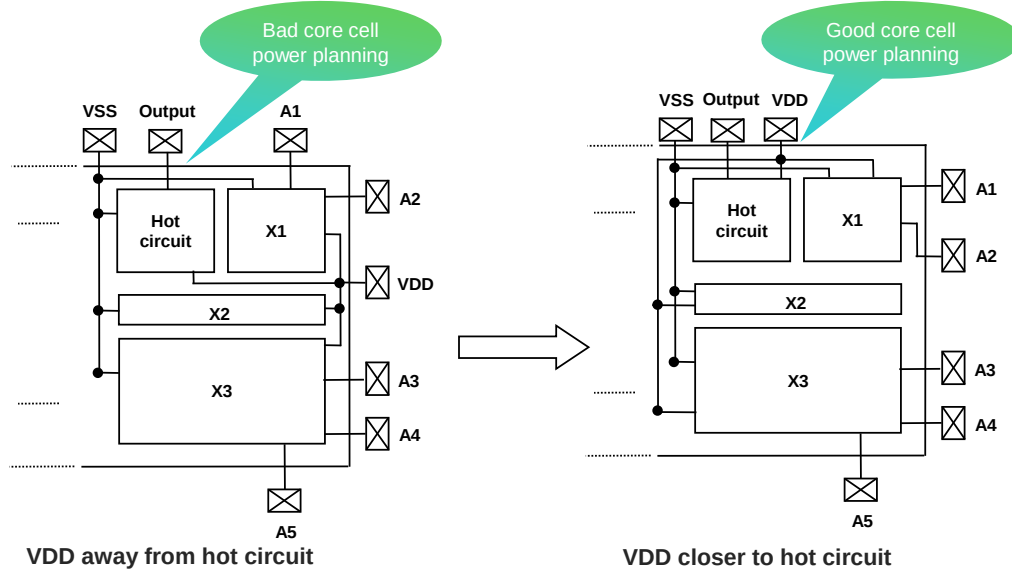
Power and Ground Planning (continued)

- Based on the current capability of the metal lines (for example, 2 mA/μm), the width of the power and ground connections should be defined.
- The location of the power and ground pads should be close to the high power cells.
- Having multiple power and ground pins helps to get power connection to high power cells more easily.



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Example: Power and Ground Planning



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Power Grid Tips

Understand the way the block either sources or sinks current.

- High-power analog blocks should have sufficient power grid over the block to have least IR drop issues on the block.
- Moderate digital logic can tolerate a lighter grid.
 - Consider the peak current consumption.
- Heavy logic switching simultaneously should be spread.
 - More metal should be added to the grid.

Try to keep power and ground lines long and wide.

- Better return path for signals.
 - Reduces loop inductance.
- Better sharing of grid metal.
 - Current flows in metals that are not used at the moment.
- Remember that an increase in length (as R increases) increases the IR drop.



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Subsection

Power and Ground Planning

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Tips for Placement

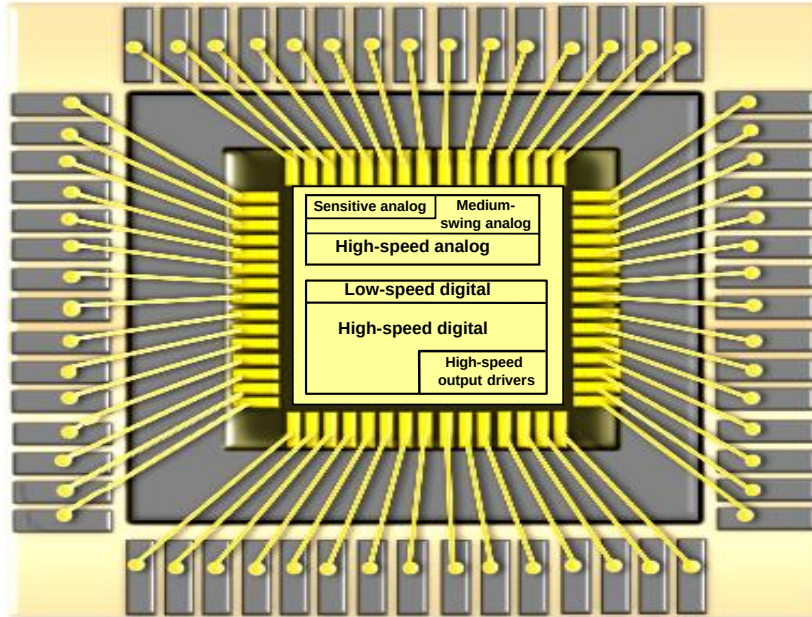
Use these tips to create an ideal floorplan.

- Place high power analog output driver circuits away from sensitive analog circuits.
- Place the high speed analog block in the middle of the die to have least stress gradient.
- Place less critical digital blocks (less in frequency or slew rate operation) near critical analog circuits to reduce noise propagation.
- Place high power output drivers near to the output pad (on the edge of the die).
- Place critical analog blocks away from high speed digital blocks.
- Let us collate all these guidelines and create a floorplan.



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Example Floorplan



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Routing

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Module Objective

In this module, you learn how to

- Create a layout from a schematic, using manual and automatic methods.



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Layout Structure

The overall structure of the layout of a specific block can be based on the following factors:

- Available size based on the floorplan
- Inputs and outputs of the block
- Matching required transistors
- Power and ground rail connection
- Shielding and sensitive node requirement



Structure of a block layout is decided WRT many factors:

- Space availability on the full chip floorplan WRT block area/aspect ratio calculation.
- I/Os of the blocks decide the aspect ratio, e.g., Output of an analog block has to connect to a huge digital bus, then the analog block is spread such a way, that a uniform parasitic bus is created to get connected to the digital block.
- Matching constraints decide the aspect ratio of the block, e.g., Highly sensitive circuits are of aspect ratio of 1.
- Top level power ground rail connections WRT block level power rails.

Device Placement Within a Block

During analog device placement, every device in the circuit is analyzed in detail for its impact on final performance of the circuit.

- Circuit performance can be affected by different factors, such as:
 - Process variation
 - Parasitic components
- To immunize the circuit layout to these factors, follow various layout guidelines to place devices efficiently during cell layout.
- Using automated cell planning, you can generate a first draft of the layout and then make improvements.



Process variation is an important factor that has to be considered during analog layout placement.

Particularly DSM (deep sub-micron <100nm) node designs are highly prone to process variants which could affect the circuit performance significantly.

Process Variation Factor

To immunize the layouts to process variations, you can use some of the following techniques:

- Device matching
 - Common centroid or interdigitated placement
 - Dummy devices
 - Orientation
- Wire matching
- Pin placement



Advantages of using analog layout techniques, are discussed detail in Module 8.

Device-Matching Guidelines

- Match devices in a common-centroid pattern:
 - Common-centroid layout adds more parasitics to the design because of the complicated routing.
 - For parasitic-sensitive design, use interdigitation techniques to match devices.
- Use a dummy device.
- The same orientation is required for matched devices.

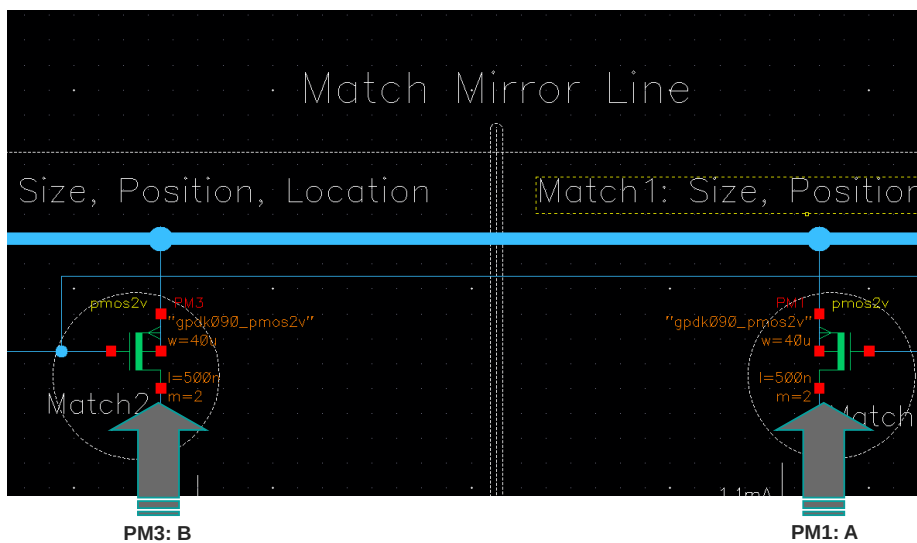
If devices in different blocks are supposed to have some kind of matching, place them in the same orientation.



Following these matching guidelines, leads to:

- Higher immune to process variations.
- Predictable circuit performance.
- Improved reliability of the circuit.

Device-Matching Guidelines (continued)

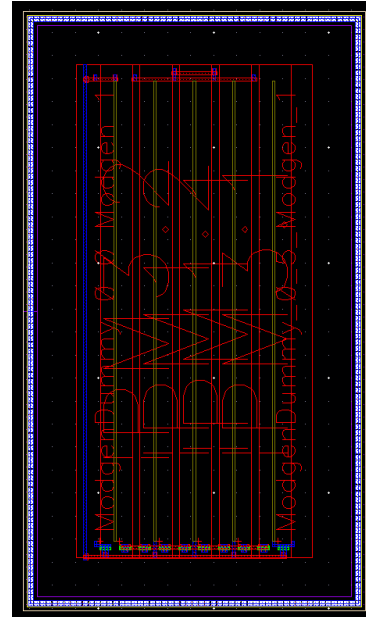


Matched devices PM3 and PM1

PM3 and PM1 are said to be the matched devices as they are the current references for this circuit.

Device-Matching Guidelines (continued)

- PM1: A
- PM3: B
- M factor: 2
- Placement pattern:
- Dummy devices are added on the ends to have higher matching gradient.
- Matching pattern:



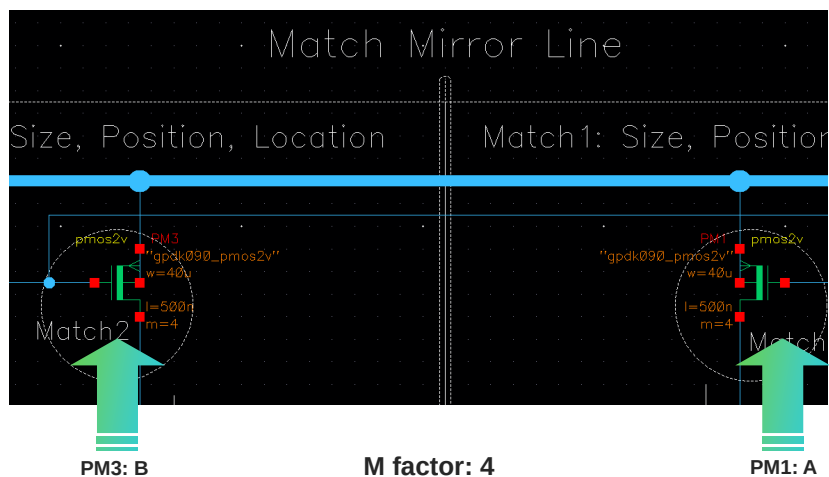
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The above said pattern is a typical common-centroid pattern which is followed for two finger configuration of two matched transistor.

Detailed discussion on device matching is discussed in Module 8.

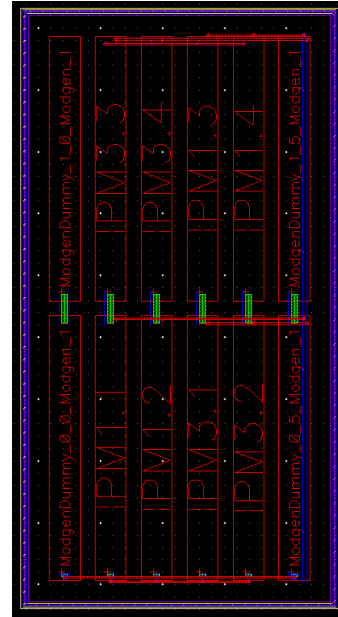
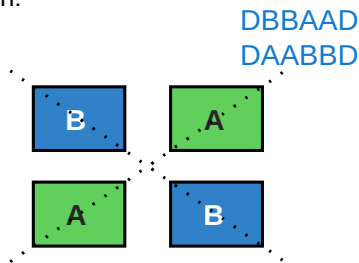
Device-Matching Guidelines (continued)



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Device-Matching Guidelines (continued)

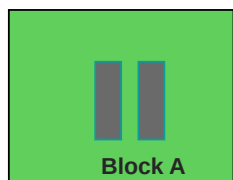
- PM1: A
- PM3: B
- M factor: 4
- Placement pattern: BBAA
AABB
- Dummy devices are added on the ends to have higher matching gradient.
- Matching pattern:



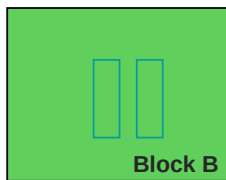
The above said pattern is a typical common-centroid pattern which is followed for four finger configuration of two matched transistor.

Detailed discussion on device matching is discussed in Module 8.

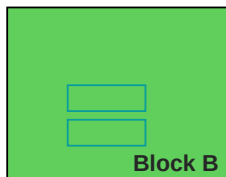
Device-Matching Guidelines (continued)



Good layout for block 2



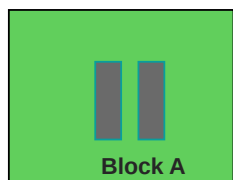
Bad layout for block 2



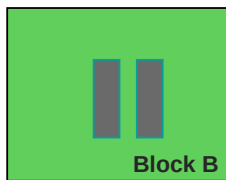
Orientation

Orientation of the critical devices and the direction of the current flow in these devices are an important factor that needs to be considered while deciding the placement of these devices.

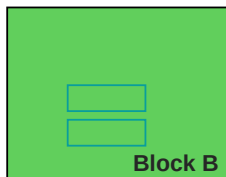
Device-Matching Guidelines (continued)



Good layout for block 2



Bad layout for block 2

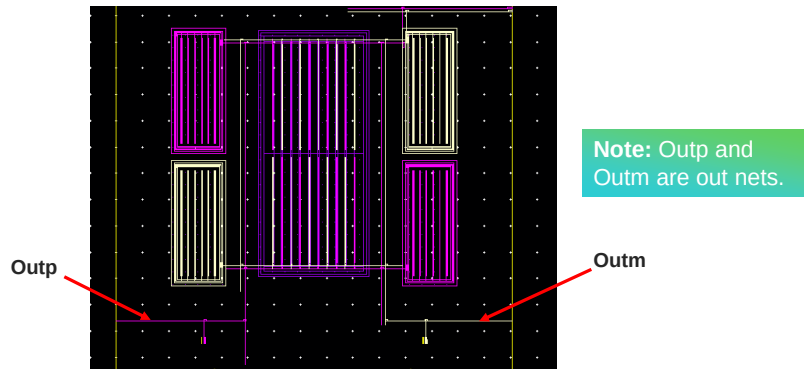


Orientation

Orientation of the critical devices and the direction of the current flow in these devices are an important factor that needs to be considered while deciding the placement of these devices.

Wire-Matching Guidelines

- Route differential signals together.
- Use symmetric connection to the matched device for differential signals.
- Use same width of the metal for differential signal routing.



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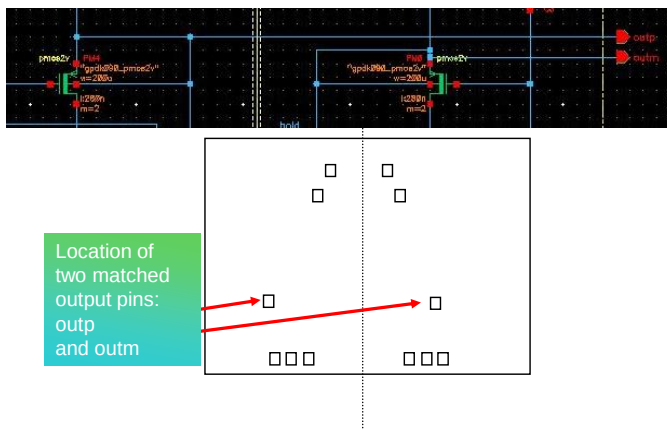


The above shown picture depicts a perfect mirror symmetry routing of two matched transistors.

This Matched mirror symmetry routing is followed for critical differential pairs transistors. These transistors need to be placed in matched fashion to achieve perfect symmetrical routing.

Pin Placement Guidelines

- Defines input and outputs
- Chose symmetric pin placement based on device placement.



Similar to symmetrical placement and symmetrical routing, even symmetrical PIN placement is also a major constraint for analog layouts.

Parasitic Component

Even after considering the parasitic effects of a layout, the layout should follow certain guidelines to improve the performance of the circuit. The layout should have:

- Lower interconnect parasitic
- Lower crosstalk
- Lesser IR drop
- Lower ground bounce
- No latchup



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Parasitic-Component Guidelines

- Define the sensitive nodes on the schematic and shield these nodes.
- Define noisy nets on the schematic and route these away from sensitive nodes to have lower crosstalk.
- Define high current path on the schematic, choose the width of the routing, and place the related device close to the associated pad.
- Define the switching part of the circuit, choose the appropriate width for its ground, and supply the connection to have less ground or supply bounce.
- Put in enough pickup for substrate and n-well to prevent latchup.



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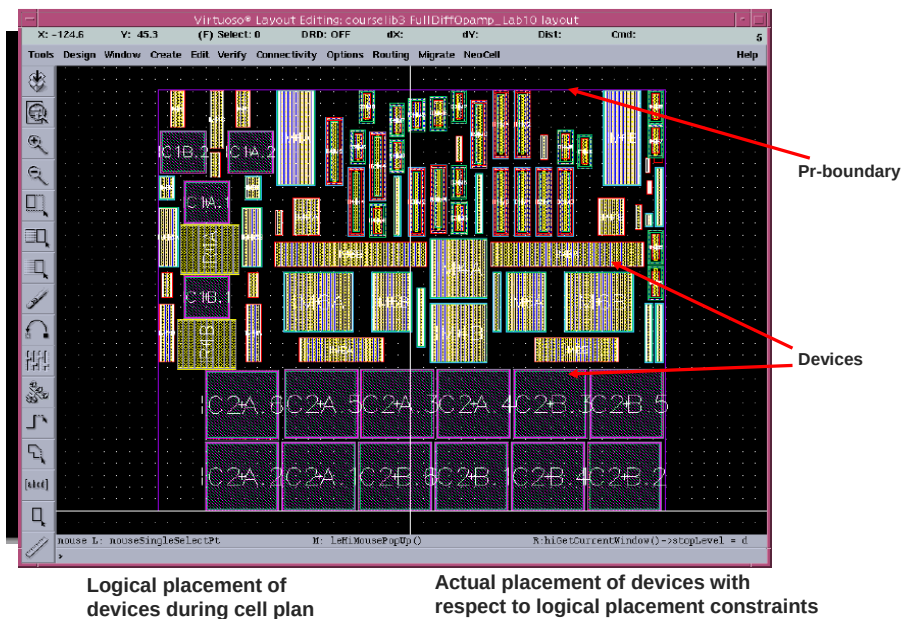
Automated Cell Planning

- The whole placement technology can be automated. It can improve the productivity of the designers, who can use some of the latest layout synthesizers that are available.
- Using Cadence's Virtuoso NeoCell, you can automate the process of defining the device and net level constraints and implement them with higher precision and faster.



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Automated Cell Planning: Example



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WRT → with respect to.

Analog Routing Steps

After the placement of devices is finalized, the following flow is used to start the custom route on analog designs:

- Pin placement
- Routing critical nets using wire editor
- Routing power grid using Virtuoso Chip Assembly Router
- Routing the remaining noncritical nets, either using Virtuoso Chip Assembly Router or Wire Editor

Most of the previous flows are done with Virtuoso® XL Wire Editor or Virtuoso Chip Assembly Router.

Wire Editor: A feature of Virtuoso XL that enables you to use interactive routing and design-rule-driven wire stitching. Wire Editor integrates the interactive Virtuoso Chip Assembly Router wire-editing capabilities into Virtuoso XL.



WE → wire editor

VCAR → virtuoso chip assembly router.

Each of the steps: Pin placement etc.. Will be explained in the next few slides.

Pin Placement

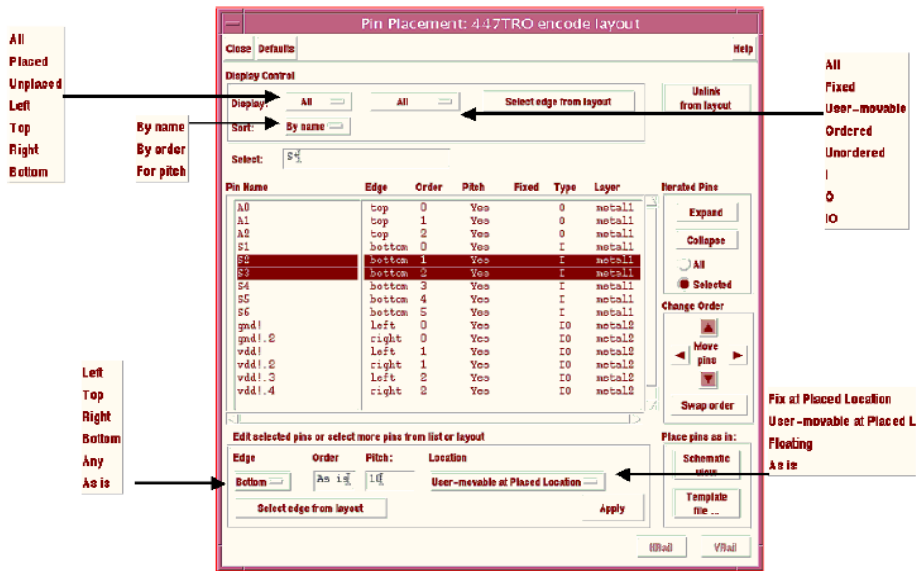
- Assigns specific ordering for a group of pins
- Places pin group with any given pin pitch spacing
- Expands or collapses iterated bus pins ($Q<7:0> = Q<7>, Q<6>, Q<5>$)
- Assigns fixed constraint on pins
- Supports template file and schematic view
- Creates rail on power and ground pins
- Updates pin placement in real time



Pin placement is an activity, which is partly decided during floorplanning and partly modified during block level designs.

There are automatic pin placement features in VXL through which you can automate the complete pin placement procedure.

Pin Placement (continued)



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VCP: Pin Placement Sub Menu.

Here you will be able to achieve the following using this menu:

- Assign specific ordering for a group of pins.
- Place pin group with any given pin pitch spacing.
- Expand/collapse iterated bus pins ($Q<7:0> = Q<7>, Q<6>, Q<5>$).
- Assign fixed constraint on pin.
- Support template file and schematic view.
- Create rail on power/ground pin.
- Update pin placement in real time.

Critical Net Routes: Wire Editor Features

- Routes interactively in Virtuoso XL with Wire Editor
- Implements real-time interactive DRC feature within Wire Editor
- Routes single or multiple nets, as well as buses based on connectivity and DRC rules
- Pushes and shoves with nets, vias, and components while routing
- Checks existing routes for violations
- Reshapes wire segments
- Eliminates notches and removes extra bends
- Compacts existing routes
- Copies existing routes to unrouted connections
- Generates reports about routing conditions and objects

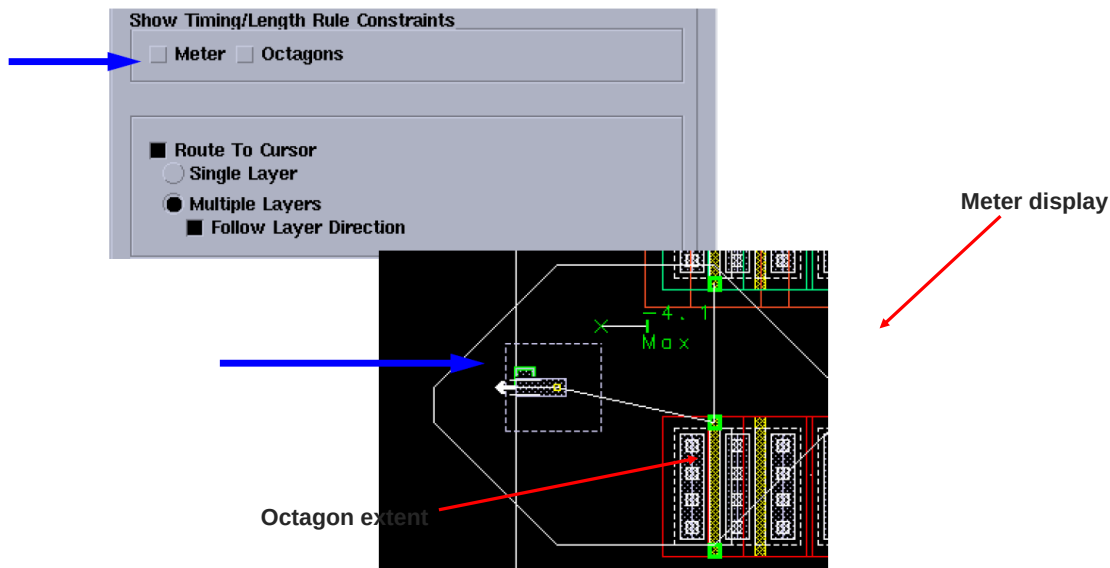
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Wire editor:

A feature of VXL that enables you to have interactive routing and design-rule-driven wire stitching. The Wire Editor integrates the interactive Virtuoso® Chip Assembly Router wire editing capabilities into Virtuoso XL.

Critical Net Routes: Wire Editor Features



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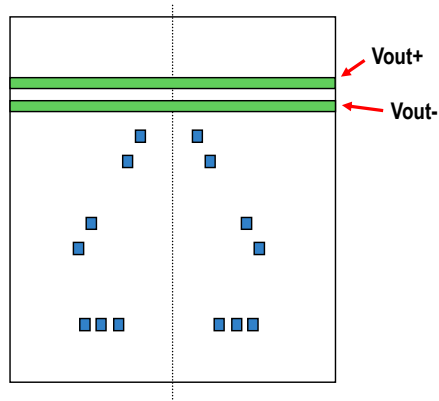


Wire editor can dynamically show the routing statistics on the particular net that we are drawing. This feature enables the user to exactly calculate the length of the drawing wire which eases the job of a designer while drawing matched wires.

- Show Timing Length/Rule Constraints displays length rule indicators while a wire with a length rule is being created.
- Meter displays a negative number in green if the wire being created is within the length rule limits. It displays a positive number in red when the wire being created is outside the limits.
- Octagon shows the extent to which a wire can be routed within the length rules.

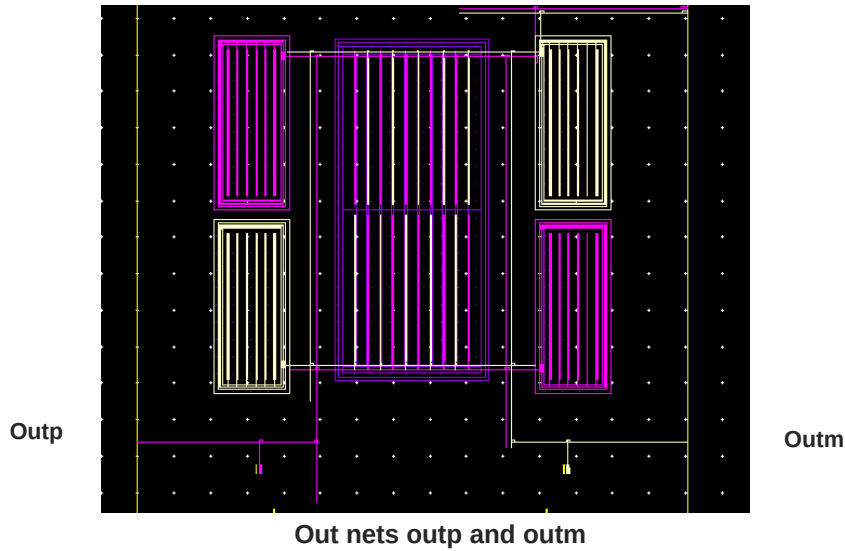
Critical Net Routes: Matched Output Nets

- Symmetrical output differential nets are kept matched to have the same output parasitics for the differential output drivers.
- These critical matched routes can be drawn using Wire Editor.



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Example: Matched Output Nets with Wire Editor



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WE editor:

Outp layout net is highlighted using purple and outm net is highlighted using yellow.

You can notice that designer has drawn a perfect matched routing using WE, here the expertise of the designer plays a major role for a high-quality layout as most of the job is done manually using WE.

Remaining Route: Virtuoso Chip Assembly Router

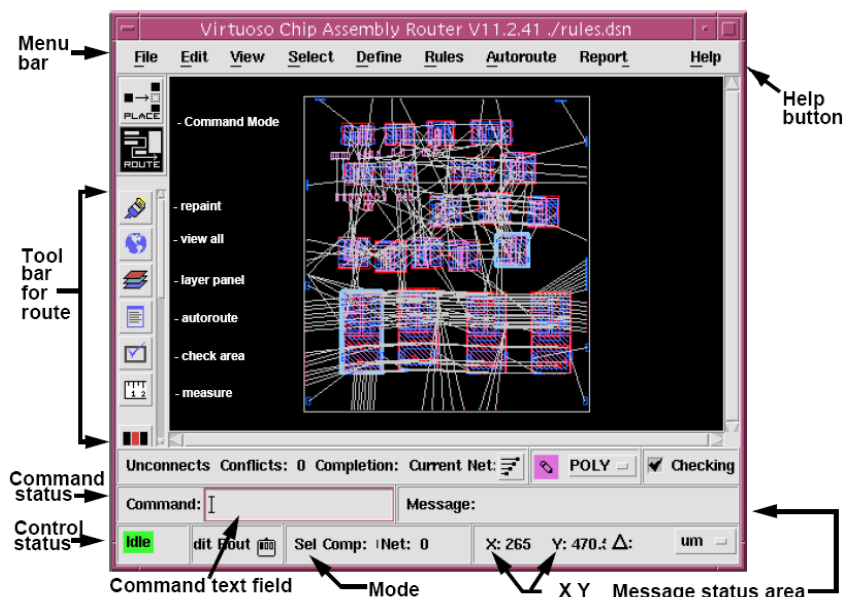
Virtuoso Chip Assembly Router is a Cadence chip assembly router that works on layout shapes rather than routing grids. Virtuoso Chip Assembly Router is capable of routing in following design styles:

- Differential pairs and net shielding
- Impedance-aware routing
- Wire segment crosstalk
- Accumulative coupled noise power trunk routing
- I/O ring and power routing
- Interactive bus routing
- Interactive topology editing
- Detail routing



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Virtuoso Chip Assembly Router GUI

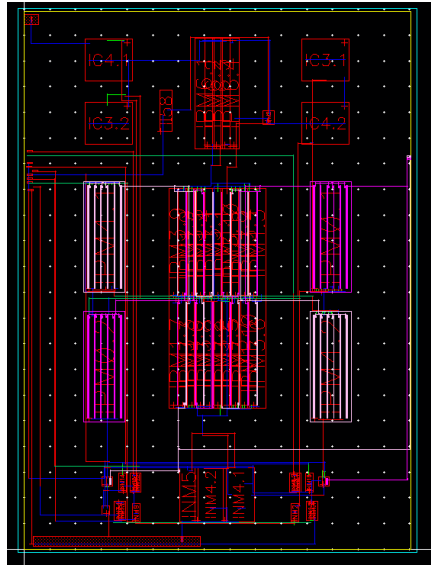


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Signal Routing: Virtuoso Chip Assembly Router

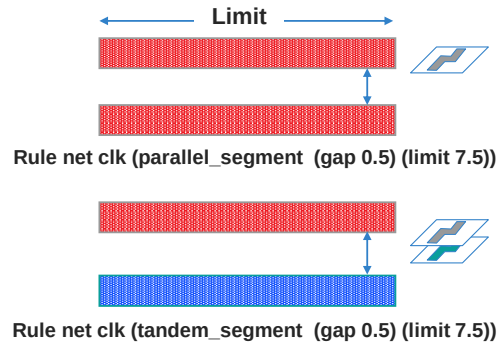


Layout cell after Virtuoso Chip Assembly Router signal routing

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Crosstalk-Aware Routing: Virtuoso Chip Assembly Router

The **parallel_segment** and **tandem_segment** rules are simple controls for the length and spacing of wire segments only. You set a segment length *limit* for each spacing value or *gap*. When the limit threshold is exceeded for any specified gap, those segments are rerouted.



VCAR has few cross talk aware routing features when it is trying to route long wire automatically.

- **parallel_segment**: With this constraint, VCAR limits the length of a parallel routed nets to a certain length limit above which tool reroutes the net to avoid cross talk between these nets.
- **tandem_segment**: this does the same as the above parallel_segment rule, but here the difference is that the limit works for parallel routes which are located in adjacent layers, this is to avoid cross talk due to interlayer capacitance.

Cell-Placement Techniques: Custom Digital

Custom digital designs usually consist of

- Transistor level: Logic standard cells
- Cell level: Data path, memory and clock blocks, and so forth

Following are some placement techniques that are used in custom digital-design styles:

- Start with the critical path.
- Place cells to limit the wire load on the critical path.
- Move less critical blocks out of the way.
- Place clock generators to limit clock wire load.
- Consider track-sharing opportunities when placing cells.
- Follow the data flow in cell placement.
- Try to follow the data-bus path while deciding the placement of memory elements.



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Area Estimation: Custom Digital

Data path blocks

- Pitch matching all the cells to match the memory pitch is a major concern during data-path block designs.
- Determine if signal or power routing is the design area limiter. Usually, adders and shifters are more routing limited than the device area.
- Routing limited: Width of block set by bit pitch; height of cells is set by signal routing.
- Nonrouting limited: Use the transistors per square microns empirical formula for calculations.

Memory blocks: As the unit memory block is constant, the total area of the block is determined by the total number of memory elements and peripherals.

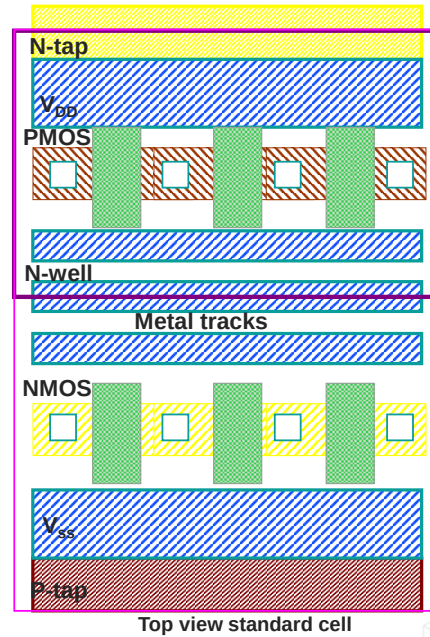
Clock blocks: Generally placed near the flops.



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Planning Layout Structure: Logic Standard Cell

- Logic standard cells always follow a fixed standard cell template for which all devices and routing should comply.
- All PMOS devices are placed inside the n-well near V_{DD} and NMOS devices are placed near V_{SS} .
- All the transistors have to be folded or fingered to fit the height of the template.
- Pins of the standard cells have to be placed on the routing tracks.
- All transistors that can be shared have to be chained or folded to have the least diffusion capacitance after placement.



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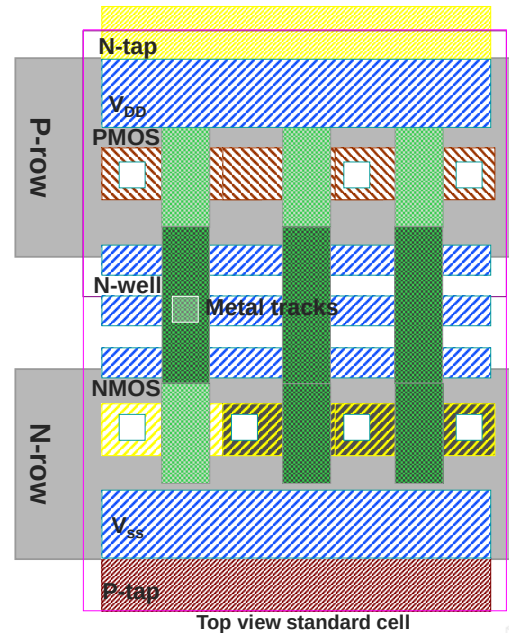
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Planning Layout Structure: Logic Standard Cell

Standard-cell templates are designed in such a way that the following can be accommodated with the stipulated height:

- Grouping of NMOS and PMOS devices under n-rows and p-rows.
- Devices get grouped using transistor chaining and folding.
- Alignment of devices are achieved for straight gate connections.
- Poly and metal1 metals are used for transistor-level connectivity.



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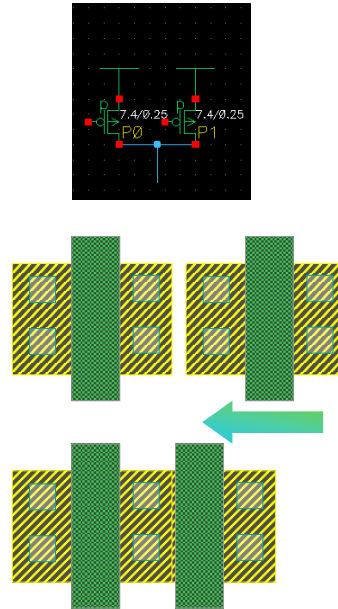
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Animation explains the sequence of the statements stated as bullets on the left side of the slide.

Device-Level Grouping: Abutment

Device-level grouping is achieved through abutment, chaining, and folding.

- Abutment allows you to share the source or drain of devices on the same net. Abutment reduces the source or drain diffusion capacitance and improves the design performance.
- Diagram shows two same-width devices with a common source are being abutted together to save space and reduce diffusion capacitance.



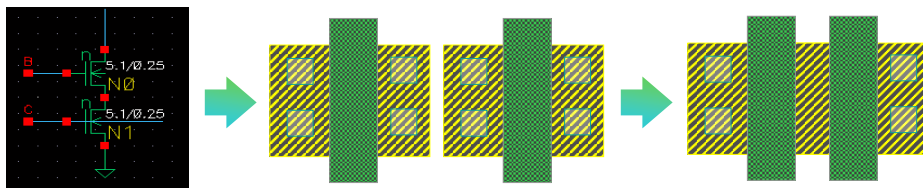
Abutment of transistors can bring down diffusion capacitance significantly.

At the same time abutment can save a lot of space consumed by a standard cell.

Device Level Grouping: Transistor Chaining

Transistor chaining: A process of sharing diffusion for series transistors with common source or drain.

- If the common source or drain net of a series transistor does not connect anywhere else in the layout, you can remove the source or drain contact to save space of the overall cell.
- By reducing the space between two gates of the series transistors, you can improve the speed of operation of these transistors.

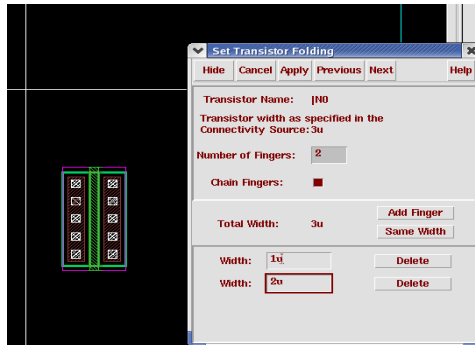


Transistor chaining is used during series transistor configurations to attain the following advantages:

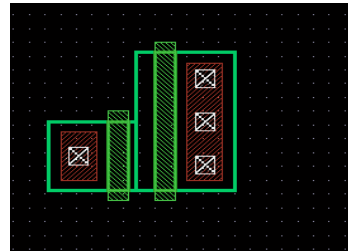
1. Reduced diffusion capacitance and resistance.
2. No diffusion routing required
3. Saves overall space of the design.

Device Level Grouping: Transistor Folding

Transistor folding: A process of splitting a long transistor to form small parallel transistors, which aid designers to accommodate the transistor within the restricted height of the standard cell.



A long transistor



Folded transistor with two fingers

Transistor folding is a technique to reduce the overall resistance of the huge device by splitting the devices to multiple parallel connected devices.

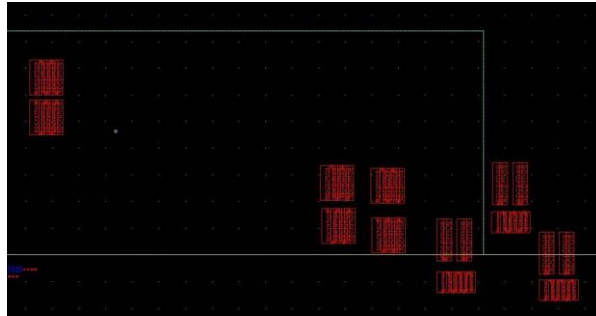
This technique is used to fit a huge transistor within the standard cell height.

Folding a huge transistor and placing them with a fingered configuration improves the performance of the circuit due to reduced device parasitic.

Planning Layout Structure: Device Level

To follow the standard cell template and circuit constraints, all PMOS and NMOS devices of the standard cell are grouped together and assembled with in n-rows and p-rows respectively. Consider the following example as shown in the figure:

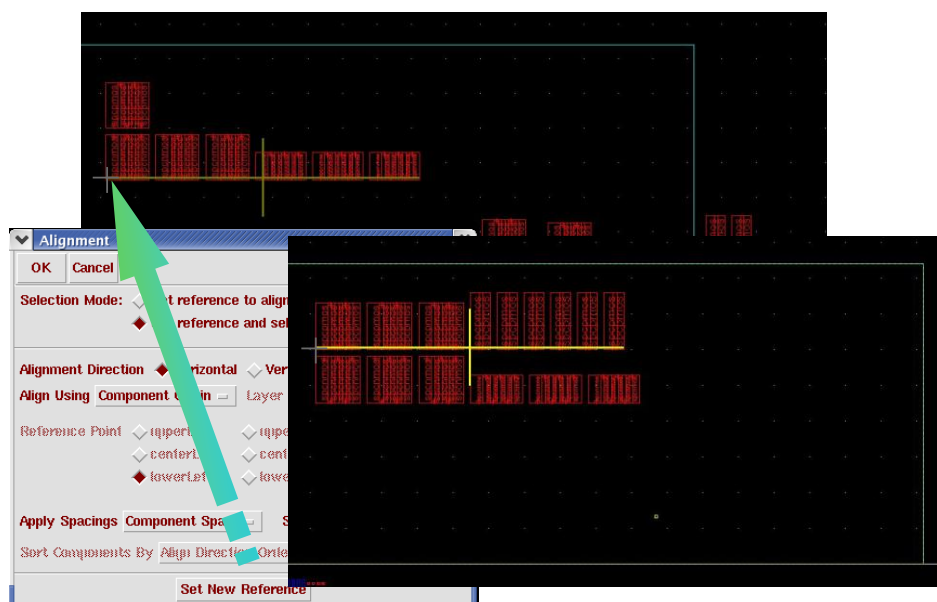
- These devices have to be arranged in a row to get them routed.
- Row-wise placement of devices can be automated through alignment command of Virtuoso XL.



Unassembled layout structure after Generate from Source command of Virtuoso XL

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Device Level Assisted Alignment



Aligning of transistors within a n/p-row is well automated using the align command of VXL.

Planning Layout Structure: Cell Level

During custom-block-level designs, all standard cells are placed in the layout environment and assembled manually with respect to the circuit constraints.

- The standard cell assembly generally follows an alternate flip placement technique (P-N-N-P or N-P-P-N) to share the n-wells of the p-devices that are present inside the standard cells.
- Generally, the smaller-node standard cells do not come with well or substrate taps because they consume more space due to DRC limitations. So, designers insert these substrate or well taps at regular intervals on the top level of the block to protect the circuit from latchup failures.
- Filler cells can be inserted manually or automatically to fill the empty spaces between the placed standard cells to have continuous power rail or n-well connectivity with the block.



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Subsection

Planning a Layout Structure – Leaf Cells (Custom Digital Layout)

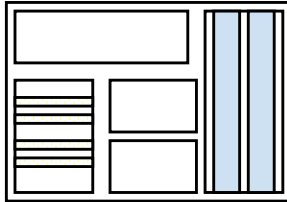
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Planning Layout Structure: Partitioning

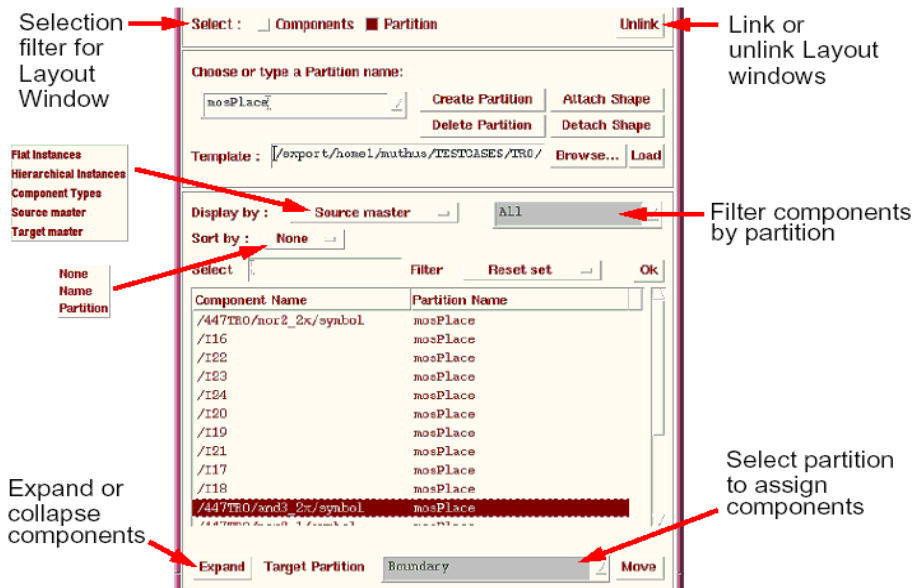
Partitioning enables users to plan the layout in more detail by clustering components together and assigning them to specific areas (partitions) of the layout, for example, analog circuit, memory subblock, and so forth.

- The design's **prBoundary** is the default partition with the keyword *Boundary*.
- You should target individual partitions for placement planning, for example, row generation and support the template file.



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Partitioning (continued)



Partitioning is an activity through which you can logically assign device placement constraint.

In the screenshot, you can create several partitions that can later be assigned to different constraints.

These partitions include:

- MOS PLACE → mos transistor level placement.
- Standard cell → standard cell level placement

Placement Planning

Assisted mode automates typical styles of MOS and standard-cell placement and helps the user to generate rows in each partition:

- Assisted CMOS
- Assisted standard cell
- Assisted mixed CMOS and standard cell

An estimator calculates design statistics in assisted mode, for example, row count, area utilization, device count, and device width.

Evaluate the design statistics to determine whether it is necessary to update constraints and regenerate rows.



Update rows in real time.

Limitations of Assisted mode are documented in cdsdoc

Placement Planning (continued)

Assisted CMOS
Assisted Standard-Cell
Assisted Mixed CMOS/Standard-Cell
Manual User-Defined

Partition: Boundary [Select From Layout]

Style: Assisted CMOS [Link]

Style File: [Browse...] [Load]

Load Row Template For: Entire Design [Selected Partition]

Template File: [Browse...] [Load]

Region: Row [Rais] [Components]

Region Shape: Rectangle [Polygon]

Origin: X [Y]

Width [Height]

Draw [Update] [Delete]

Properties:

☐ Allow For Pins ☒ Horizontally

☐ Allow Region To Expand ☒ Vertically

☐ Allow Partition To Expand

Layout Generation: Regenerate All [Preserve Constrained Objects] [Generate Only Missing Components]

Estimates: Number Of Rows: 0

Region Utilized (%):	Count	Within Row			Between Row		
		Max. Width	Min. Width	Avg. Width	Max. Width	Min. Width	Avg. Width
PMOS:	0	0.0	0.0	0.0	0.0	0.0	0.0
NMOS:	0	0.0	0.0	0.0	0.0	0.0	0.0

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- Separate tabs for region, row, rail, and component attributes.
- Set placement grids within standard cell and mixed row definitions.
- Control chaining, stacking, and chain size parameters from within the Placement Planning form.
- Mixed mode allows placing CMOS and standard cell components on the same row.

Placement Planning (continued)

- Manual/user-defined mode offers greater flexibility but less automation. If any limitation in assisted mode is not acceptable to the user, use manual/user-defined mode instead.
- No estimator is used in manual mode. The user has full control and can encounter unexpected overutilization.
- In manual mode, the user manually draws rows and defines the component type used for each row.
- Tools generally recommend to use assisted mode as much as possible and then refine in manual mode.



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Placement Planning: Standard Cells

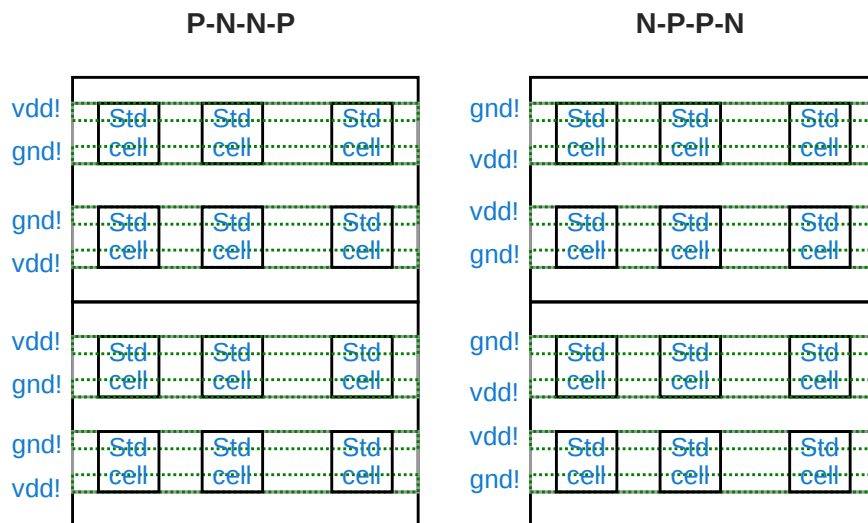
Requirements for standard cell libraries to achieve optimal placements.

- Correct by construction, that is, all standard cells are designed to be abutted without creating design rule errors between each standard cell.
- Each standard cell should have cell boundary in **prBoundary dg** or **prBoundary by** or **instance dg**. If none exists, Virtuoso Custom Placer tries to derive one, but it might not be optimal.
- Power rails of all the standard cells can be placed differently:
 - P-N-N-P
 - N-P-P-N
 - P-N
 - N-P



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Power Rail Assembly Technique:



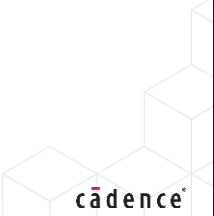
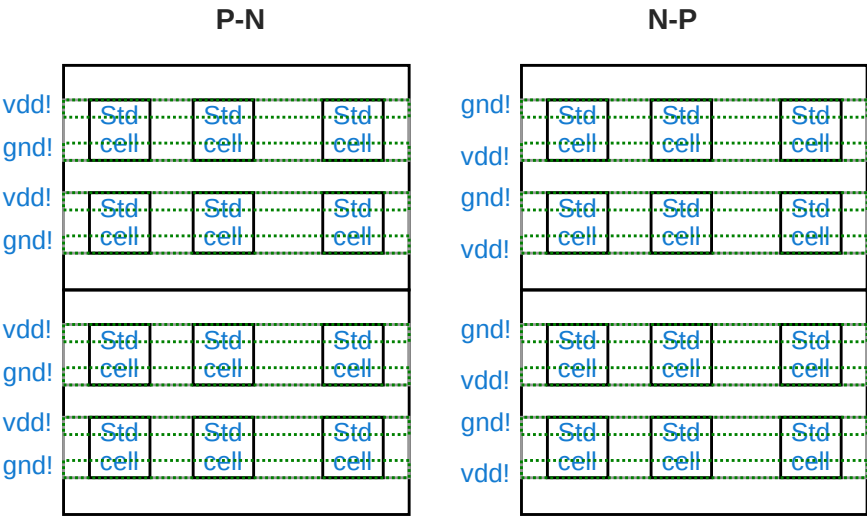
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Power rail assembly styles have the following advantages:

- We can share the n-wells of p-rows of inverted standard cell configuration.
- Common place for placing well-taps and substrate taps.
- Improving the power delivery by inserting many vias between two adjacent power rails.

Power Rail Assembly Technique (continued)



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Power Rail Assembly Technique (continued)

The screenshot shows a dialog box titled "Power Rail Assembly Technique" with four tabs: "Region", "Row", "Rails", and "Components". The "Rails" tab is selected. The dialog contains the following fields and options:

- Power Rail:**
 - Layer:** A dropdown menu showing "metal1" and "dg".
 - Width:** A text input field with "1.0".
 - Net Name:** A text input field.
- Ground Rail:**
 - Layer:** A dropdown menu showing "metal1" and "dg".
 - Width:** A text input field with "1.0".
 - Net Name:** A text input field.
- Rail Pattern:** A dropdown menu showing "P-N".
- Position:** A dropdown menu showing "Outside".
- Row Spacing Within N-P Pair:** A text input field.
- Minimum N-P Component Spacing:** A text input field.

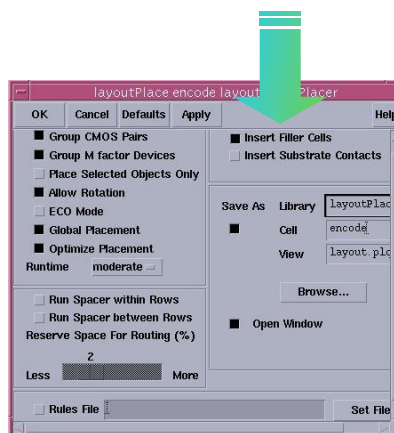
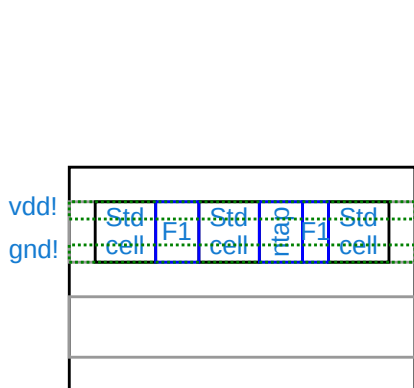
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Rails is a subsection of placement planning window. Here you can decide on the following WRT power rails:

1. Power rail style.
2. Power line metal.
3. Frequency of the power rail distance.

Filler Cell Insertion



Automatic insertion of filler cells reduce the time of manually inserting filler cells by the user.

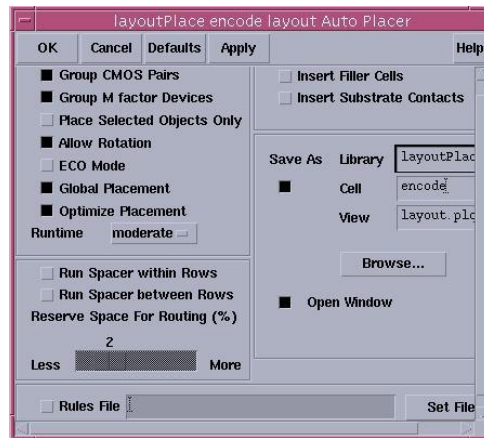
Placer Features

- Group devices based on logical information (CMOS pair, M factor).
- ECO support.
- Various levels of optimization; global placement alone can provide good result.
- Reserve space for routing.
- Options to insert filler cells and substrate contacts.



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Placer Menu



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Once you have defined the placement style and areas, run the autoplacer with the command **Place—Placer** from the VXL Layout Editor pull-down menu.

Preserve Logical Hierarchy

To maintain grouping of devices per the schematic hierarchy. This is a very useful option for placement of gate level cells (inverters, nands, nors, etc.).

Place Selected Objects Only

Partitions your design with selected instances for placement.

Allow Rotation

Can override defined allowed rotations and temporarily fix rotations for that placement run.

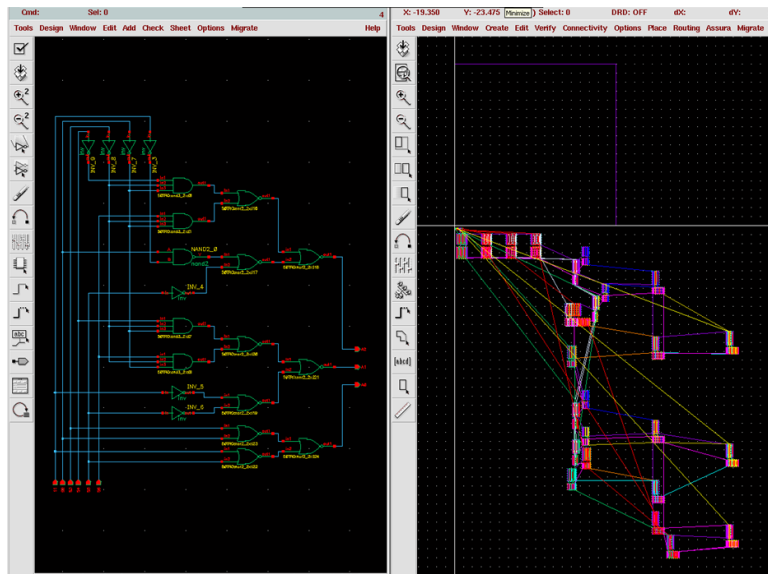
ECO Mode

Select ECO Mode for incremental updates only. It will not disturb anything it doesn't have to and adds new devices as needed. Will not preserve logical hierarchy.

Global Placement and Optimize Placement

Always run Global before any detailed placement passes. Global places components without regard to their initial placement, as overlap violations between components might be ignored. The placer optimizes global objectives such as total wire length. Typically, you would select this option the first time you place a design or to discard unsatisfactory placement results. Use Optimize until objective cost is achieved.

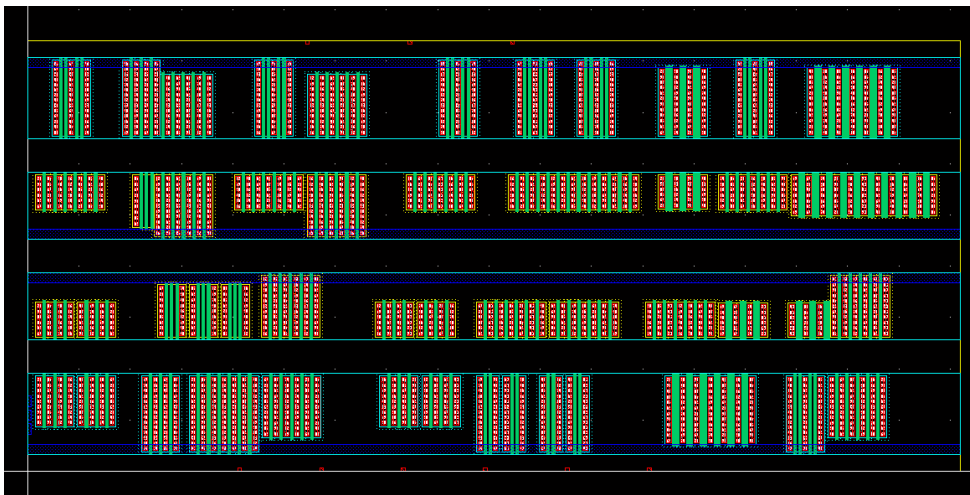
Virtuoso Custom Placer Placed Design: Example



Layout generation from schematic using Generate from Source command

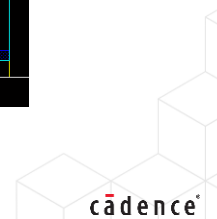
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Virtuoso Custom Placer Placed Design: Example



Fully assembled block using automatic placement
(Virtuoso Custom Placer)

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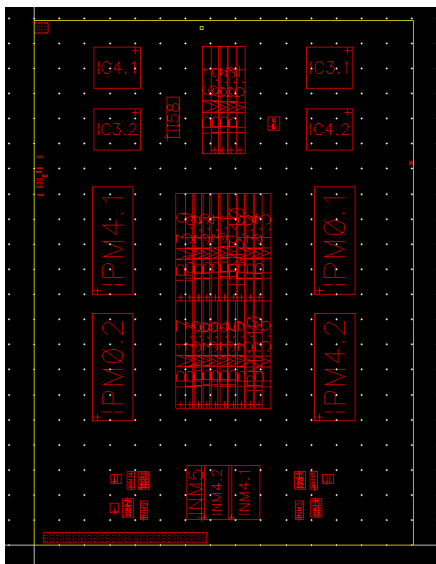


The above graphic is an output of the automatic placer, VCP (Virtuoso Custom Placer).

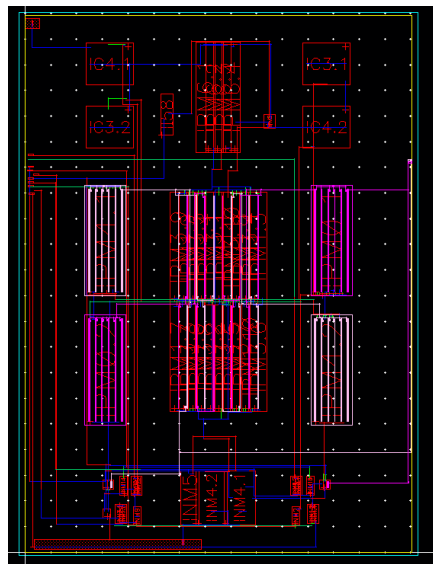
The cell was assembled in this fashion:

- Pin placer was used to decide the pin locations.
- Power rails were drawn automatically using the pin placer.
- Partitioning and assigning on devices to the partition was done.
- Partition planning was done.
- VCP was run last.

Device Level Routing: Virtuoso Chip Assembly Router



Layout cell before Virtuoso Chip Assembly Router signal routing



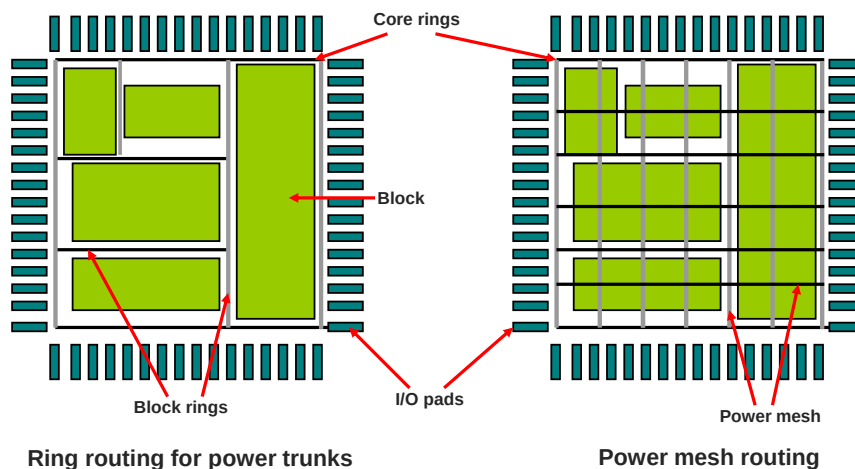
Layout cell after Virtuoso Chip Assembly Router signal routing

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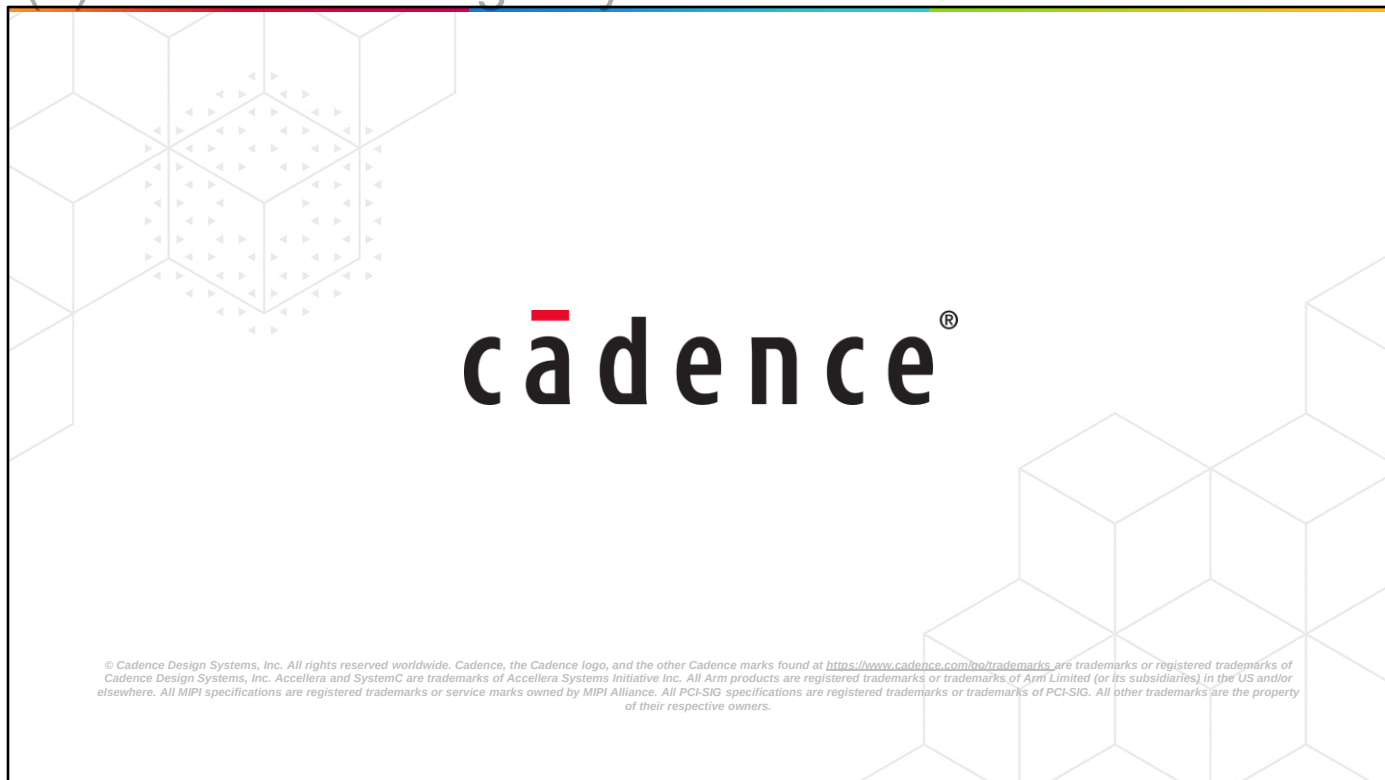


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Power Routing: Virtuoso Chip Assembly Router



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